Finite-Step Sinkhorn Certificates for Nonnegative Stream-Mixing Products

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Abstract

We give deterministic finite-step Sinkhorn certificates for the nonnegative stream-mixing pathway in expanded-stream residual architectures, motivated by Manifold-Constrained Hyper-Connections (mHC). The central object is the depth-wise mixer product

$$T_L = H_{\text{res}}^{(L)} H_{\text{res}}^{(L-1)} \cdots H_{\text{res}}^{(1)},$$

not an unconditional stability theorem for the full nonlinear network.

For entrywise nonnegative mixers, row and column marginal errors are induced-norm certificates: they bound the forward ℓ_∞ and backward ℓ_1 streamwise $A_{\max}$ gains. Thus finite Sinkhorn steps give a theorem chain

$$\varepsilon_c^{(\ell)} \Rightarrow \|H_{\text{res}}^{(\ell)}\|_1 \le 1 + \varepsilon_c^{(\ell)} \Rightarrow \|T_L\|_1 \le \prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}).$$

For the `column-then-row` normalization order considered here, we prove a Hilbert-projective finite-step bound with the correct half-step loss for the output column error. This maps logit dynamic range and Sinkhorn step budget to a certified depth-wise backward budget for the mixer product.

We also separate certificates from predictors. Worst-case Hilbert bounds are safe but pessimistic; near the Sinkhorn limit, the sharp local full-step rate is $\sigma_2(P)^2$, where P is the limiting doubly stochastic scaling. We state a conditional local derivative-envelope certificate and first-/second-order finite-step predictors for interpreting observed decay curves. The numerical section is diagnostic: it reports achieved marginal errors, product certificates, exact realized backward gains by vector propagation, cap failures, and the gap between worst-case certificates and local predictors.

1 Introduction

Deep neural networks rely on controlled signal propagation to avoid vanishing or exploding gradients. Standard stabilizers such as normalization layers, e.g. LayerNorm (Ba et al., 2016), and careful initialization reduce feature-scale drift, but they do not directly constrain another source of instability that appears in expanded-stream architectures: *products of stream-mixing matrices* across depth. Related work includes stability-motivated architectures (Haber and Ruthotto, 2018) and continuous-depth viewpoints (Chen et al., 2018).

In expanded-stream residual designs, each layer introduces a mixer $H_{\text{res}}^{(\ell)} \in \mathbb{R}^{n \times n}$ that routes information across n parallel streams.[1] If the depth-wise product of these mixers has

[1] Throughout, the row index is the *stream* index (not batch); batch is treated as an outer dimension and omitted from notation.

a large induced norm, signals (and, in backpropagation, gradients) can be amplified through the mixing pathway even when per-feature normalization is present.

Row-stochasticity alone does not control the backward mixing-path gain. For any $n \geq 2$, the row-stochastic matrix $H = \mathbf{1}\mathbf{e}_1^\top$ satisfies $\|H\|_\infty = 1$ but has column sum n in the first column, hence $\|H\|_1 = n$. Thus, row-only normalization can induce backward amplification on the order of the expansion rate. Finite-step Sinkhorn targets both marginals; we quantify how many Sinkhorn steps suffice to certify a prescribed depth-L backward budget.

The motivating architecture class is Manifold-Constrained Hyper-Connections (mHC), which enforce entrywise nonnegativity and drive mixers toward the Birkhoff polytope of doubly stochastic matrices. Since exact projection is computationally expensive, the mHC-motivated mixer parameterization considered here exponentiates logits to enforce strict positivity and then applies a *finite* number of Sinkhorn–Knopp normalizations (Sinkhorn and Knopp, 1967; Knight, 2008), creating a gap between the ideal (exact marginals) and the realized finite-step mixers. This paper addresses the corresponding calibration task: given depth L and a target backward budget B_1 for the mixing pathway (i.e., $\|T_L\|_1 \leq B_1$), we give a conservative rule that maps (i) logit range/temperature (dynamic range κ) and (ii) a finite Sinkhorn step budget $t_{\max}$ to a certified column-marginal tolerance that is sufficient to guarantee the depth-L budget *during training*.

The practical output is not only a post-hoc bound, but a controller for finite-step normalization. Given a target budget B_1, the theory gives: an adaptive stopping rule that redistributes the remaining log-budget across layers; a temperature–clipping condition that translates $(\beta, R, t_{\max})$ into a guaranteed tolerance; a certificate-aware regularizer that can be added to the training objective; and an exact realized backward-gain diagnostic computed by a single vector propagation. Thus the number of Sinkhorn steps is tied to a verifiable stability budget rather than treated only as a heuristic hyperparameter.

Sinkhorn–Knopp scaling and matrix balancing are classical (Sinkhorn and Knopp, 1967; Franklin and Lorenz, 1989; Knight, 2008), and Hilbert's projective metric provides a standard contraction framework for positive maps (Birkhoff, 1957; Bushell, 1973). In machine learning, Sinkhorn iterations are a standard primitive for entropic regularized optimal transport and "Sinkhorn distances" (Cuturi, 2013; Peyré and Cuturi, 2019), and doubly-stochastic attention has been pursued via Sinkhorn-style normalization (Sander et al., 2022) and alternative objectives (Shahbazi et al., 2025). Prior work often uses Sinkhorn normalization as a layer or operator; here we certify the finite-step normalization error needed to control the depth-wise product of nonnegative stream mixers.

In this paper we analyze the *finite-step* normalization regime in residual backbones and derive deterministic worst-case certificates for depth-wise products of mixers. To keep the guarantees aligned with the mixing pathway in backpropagation, we phrase them in terms of coordinate-wise max-norm (Amax) amplification. Because backpropagation through the mixing map applies $H^\top$, the induced norms $\|\cdot\|_\infty$ (forward) and $\|\cdot\|_1$ (backward) provide the natural worst-case guarantees for this diagnostic. Spectral-norm control would require norm conversions (e.g., $\|v\|_2 \leq \sqrt{n}\|v\|_\infty$) that introduce dimension factors and obscure depth-wise budgets. For depth L, define the composite mixer

$$T_L := H_{\text{res}}^{(L)} H_{\text{res}}^{(L-1)} \cdots H_{\text{res}}^{(1)}. \tag{1.1}$$

In the idealized `column-then-row` normalization order considered here, the output has exact row sums; in implementations with denominator stabilizers the row defect is measured explicitly and is negligible in the reported runs. Thus, for the idealized nonnegative mixer $H \geq 0$ one has $\|H\|_\infty = 1$, and the remaining design question is control of the backward gain via $\|T_L\|_1$ (Lemma 4.1 and Remark 4.2).

Scope of the guarantees. The certificates in this paper are mixer-path certificates. More precisely, they control the depth-wise product T_L in the induced ℓ_∞ and ℓ_1 matrix norms, equivalently in the forward and backward streamwise Amax gains of the residual mixing pathway. They should not be read as unconditional end-to-end stability theorems for the full neural network. When the nonlinear residual branch is included, one also needs independent bounds, estimates, or regularization of the corresponding branch Jacobians. We record such a composition bound below, but the main deterministic contribution is the finite-step certificate for the nonnegative Sinkhorn-normalized mixer product.

What is not claimed. We do not claim state-of-the-art empirical accuracy, nor do we prove unconditional full-network stability. The deterministic guarantee is for the nonnegative stream-mixing product produced by finite-step Sinkhorn normalization. The experiments are therefore reported as numerical diagnostics for the certificate mechanism rather than as a broad benchmark study.

Contributions.

- **Mixer-product Amax certificates from marginal errors.** For entrywise nonnegative mixers, row and column marginal errors directly control $\|H\|_\infty$ and $\|H\|_1$, giving depth-wise forward and backward Amax budgets for T_L (Proposition 3.1). We also record the exact realized backward-gain diagnostic by vector propagation (Remark 3.2).
- **Finite-step Sinkhorn calibration for the column-then-row order.** Applying Hilbert-metric contraction to the column-then-row normalization schedule, we derive a column-marginal estimate with the proper one-half-step loss at the output. This maps $(\kappa, t_{\max})$ to a certified tolerance and then to a depth-wise backward budget on $\|T_L\|_1$ (Theorems 4.7 and 4.15).
- **Matrix-specific local analysis and predictors.** For a fixed positive matrix, the local full-step Sinkhorn rate is $\sigma_2(P)^2$, where P is the limiting doubly stochastic scaling. We also distinguish rigorously certified local derivative-envelope bounds from non-certified first- and second-order predictors used to interpret numerical decay curves (Theorems 4.10 and 4.11).
- **Diagnostic implementation.** We convert the certificates into practical diagnostics and controls: adaptive global-budget early stopping, temperature–clipping calibration, a certificate-aware training penalty that encourages low-budget layers, exact realized-gain measurement, and a bookkeeping composition bound for adding independent nonlinear-branch Jacobian bounds.

Finite-step Sinkhorn certificate workflow. Given depth L, expansion n, and a target backward Amax budget B_1:

1. **Choose a log-budget.** Set $b_1 = \log B_1$. A uniform allocation gives $\varepsilon_c = B_1^{1/L} - 1$; the adaptive controller in Lemma 4.16 reallocates the remaining budget layer by layer.
2. **Choose temperature and clipping.** If logits satisfy $C_{ij} \in [-R, R]$ and $A_{ij} = \exp(\beta C_{ij})$, then $\kappa(A) \le \exp(2\beta R)$ and $q \le \tanh(\beta R)$.
3. **Set the step cap.** Choose $t_{\max}$ so that the conservative certificate or the temperature–clipping rule meets the current tolerance.
4. **Stop and certify online.** Measure the realized column-marginal error after each full step, stop once the allocated tolerance is met, and record the exact realized backward gain by vector propagation.
5. **Optional training penalty.** Add the certificate-aware penalty in Subsection 4.10 when one wants the optimizer to avoid high-budget layers.

Figure 1: Calibration from a depth-L mixer-path backward budget to clipping, temperature, finite-step Sinkhorn stopping, and realized-gain diagnostics.

Claim-to-evidence map.

Claim	Supporting result
Nonnegative marginal errors control forward and backward mixer gains	Proposition 3.1
Finite column-then-row Sinkhorn steps certify output column tolerance	Theorem 4.7
Layerwise column tolerances give a depth-wise backward budget	Theorem 4.15
Adaptive stopping preserves a global log-budget when no layer hits the cap	Lemma 4.16
The sharp local full-step rate near the Sinkhorn limit is $\sigma_2(P)^2$	Theorem 4.10
Experiments illustrate conservatism and realized gains	Section 5

Why finite-step certificates matter for trainable modules. Sinkhorn normalization is often inserted into neural architectures as a practical operator with a fixed iteration count. Without a finite-step certificate, that iteration count is only a heuristic knob: it may improve marginal balance, but it does not say what depth-wise amplification budget has been enforced. The point of the results below is to turn the finite number of normalization steps used during training into a checkable mixer-path budget, with measured failures, realized gains, and adaptive stopping rules tied to the same certificate quantity.

Implementation pseudocode for the certificate bookkeeping is given in Appendix A.

2 Preliminaries and notation

Notation summary

Symbol	Meaning
n	number of streams / expansion rate
C	feature width per stream
$H_{\text{res}}^{(\ell)}$	stream-mixing matrix at layer ℓ
T_L	depth-wise mixer product $H_{\text{res}}^{(L)} \cdots H_{\text{res}}^{(1)}$
$A = S^{(0)}$	positive pre-normalized matrix before Sinkhorn scaling
T_c, T_r	column and row normalization maps
$S^{(s)}$	matrix after s alternating normalization half-steps
t	number of full column-then-row Sinkhorn steps; output is $S^{(2t)}$
$\varepsilon_r, \varepsilon_c$	row and column marginal errors
κ	entrywise dynamic range $\max_{ij} A_{ij} / \min_{ij} A_{ij}$
$\Delta(A)$	projective diameter of A
$\tau(A)$	Birkhoff contraction factor $\tanh(\Delta(A)/4)$
q	dynamic-range bound $(\kappa - 1)/(\kappa + 1)$ for $\tau(A)$
P	limiting doubly stochastic Sinkhorn scaling of A
ρ_*	local full-step rate $\sigma_2(P)^2$
Amax	streamwise coordinate maximum functional

2.1 Streams-as-rows convention

Fix an expansion rate $n \geq 1$ and feature width $C \geq 1$. At depth ℓ, the state is a matrix $x_\ell \in \mathbb{R}^{n \times C}$ whose *rows* index streams and whose columns index features. A hyper-connected residual layer has the form

$$x_{\ell+1} = H_{\text{res}}^{(\ell)} x_\ell + (H_{\text{post}}^{(\ell)})^\top F(H_{\text{pre}}^{(\ell)} x_\ell, W_\ell), \tag{2.1}$$

where $H_{\text{res}}^{(\ell)} \in \mathbb{R}^{n \times n}$ mixes streams, $H_{\text{pre}}^{(\ell)}, H_{\text{post}}^{(\ell)} \in \mathbb{R}^{1 \times n}$ are pre/post aggregation row vectors, and $F(\cdot, W_\ell) : \mathbb{R}^{1 \times C} \to \mathbb{R}^{1 \times C}$ is a nonlinear block (e.g., MLP or attention) applied to the aggregated features. Our stability focus is the *stream mixer product* generated by $H_{\text{res}}^{(\ell)}$ across depth.

2.2 Birkhoff polytope and approximate doubly stochasticity

The Birkhoff polytope of doubly stochastic matrices is

$$\mathcal{B}_n := \left\{ P \in \mathbb{R}^{n \times n} : \; P_{ij} \geq 0, \; \sum_{j=1}^{n} P_{ij} = 1 \; \forall i, \; \sum_{i=1}^{n} P_{ij} = 1 \; \forall j \right\}. \tag{2.2}$$

Since mHC uses a finite number of Sinkhorn–Knopp steps, we quantify closeness to $\mathcal{B}_n$ via marginal errors.

Definition 2.1 (Approximate doubly stochasticity (nonnegative))**.** Let $H \in \mathbb{R}^{n \times n}$ be entrywise nonnegative. We say H is $(\varepsilon_r, \varepsilon_c)$-*doubly stochastic* (also called *bistochastic*) if

$$\max_i \Big| \sum_j H_{ij} - 1 \Big| \leq \varepsilon_r, \qquad \max_j \Big| \sum_i H_{ij} - 1 \Big| \leq \varepsilon_c. \tag{2.3}$$

2.3 Amax gains and induced ℓ_∞ and ℓ_1 matrix norms

For $M \in \mathbb{R}^{n\times n}$ we use the matrix norms induced to the vector ℓ_∞ and ℓ_1 norms:

$$\|M\|_\infty := \max_i \sum_j |M_{ij}|, \qquad \|M\|_1 := \max_j \sum_i |M_{ij}|. \tag{2.4}$$

Throughout, $\|\cdot\|_1$ and $\|\cdot\|_\infty$ refer to these *matrix* norms; when we need standard vector norms we write $\|\cdot\|_{1,\mathrm{vec}}$ and $\|\cdot\|_{\infty,\mathrm{vec}}$ (see Appendix E).

Definition 2.2 (Streamwise Amax functional)**.** For $X \in \mathbb{R}^{n\times C}$ (streams as rows), define the streamwise Amax statistic

$$\mathsf{Amax}(X) := \max_{1\le i\le n} \|X_{i,:}\|_{\infty,\mathrm{vec}} = \max_{i,c} |X_{ic}|. \tag{2.5}$$

Elementary Amax inequalities. For $H \in \mathbb{R}^{n\times n}$ and $X \in \mathbb{R}^{n\times C}$,

$$\mathsf{Amax}(HX) \le \|H\|_\infty \,\mathsf{Amax}(X). \tag{2.6}$$

Similarly, backpropagation through the stream-mixing map uses the adjoint $G \mapsto H^\top G$, and

$$\mathsf{Amax}(H^\top G) \le \|H\|_1 \,\mathsf{Amax}(G). \tag{2.7}$$

Both estimates are immediate from the triangle inequality and the definitions in (2.4).

In the streams-as-rows convention, left-multiplication $x \mapsto Hx$ mixes streams by linear combinations of rows. The *forward* Amax gain is governed by $\|H\|_\infty$, and the corresponding *backward* Amax gain is governed by $\|H\|_1$. When $H \ge 0$, the absolute values in (2.4) are redundant and these norms reduce to maximum row sum and maximum column sum.

3 Depth-wise Amax certificates

This section converts per-layer marginal errors into worst-case bounds on forward and backward Amax gains for the depth-wise mixer product. The key point is that for entrywise nonnegative mixers, marginal control directly bounds the induced ℓ_∞ and ℓ_1 matrix norms, with no additional structural assumptions.

If $H \ge 0$ is $(\varepsilon_r, \varepsilon_c)$-doubly stochastic in the sense of Definition 2.1, then

$$\|H\|_\infty \le 1 + \varepsilon_r, \qquad \|H\|_1 \le 1 + \varepsilon_c, \tag{3.1}$$

since for nonnegative H the norms reduce to maximum row sum and maximum column sum.

Proposition 3.1 (Depth-wise mixer certificate)**.** *Let $H_{\mathrm{res}}^{(1)}, \dots, H_{\mathrm{res}}^{(L)} \in \mathbb{R}^{n\times n}$ be entrywise nonnegative and define T_L as in Equation (1.1). Assume each $H_{\mathrm{res}}^{(\ell)}$ is $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$-doubly stochastic. Then*

$$\|T_L\|_\infty \le \prod_{\ell=1}^{L} (1 + \varepsilon_r^{(\ell)}), \qquad \|T_L\|_1 \le \prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}). \tag{3.2}$$

In particular, marginal error certificates propagate multiplicatively across depth into a depth-wise Amax budget for the mixer pathway.

Proof. By submultiplicativity of induced norms and (3.1), $\|T_L\|_\infty \le \prod_{\ell=1}^{L} \left\|H_{\mathrm{res}}^{(\ell)}\right\|_\infty \le \prod_{\ell=1}^{L}(1+\varepsilon_r^{(\ell)})$, and similarly for $\|\cdot\|_1$. □

Remark 3.2 (Exact backward gain by vector propagation). For realized nonnegative mixers $H^{(1)},\dots,H^{(L)}$, set $T_L = H^{(L)}\cdots H^{(1)}$. The exact backward mixing gain is obtained without forming T_L:

$$w_L = \mathbf{1}, \qquad w_{\ell-1} = (H^{(\ell)})^\top w_\ell, \qquad \ell = L, L-1, \dots, 1. \tag{3.3}$$

Then

$$w_0 = T_L^\top \mathbf{1}, \qquad \|T_L\|_1 = \|w_0\|_{\infty,\mathrm{vec}}. \tag{3.4}$$

Indeed, for a nonnegative matrix the induced ℓ_1 norm is the maximum column sum. This diagnostic is often much sharper than the layerwise product surrogate.

3.1 From mixer-path certificates to full-block certificates

The product estimates above isolate the pure stream-mixing pathway. For applications one also needs to know how the certified mixer interacts with the nonlinear residual branch. The following elementary composition bound records this coupling directly in the same Amax functional.

Proposition 3.3 (Full-block Amax composition bound)**.** *Consider a depth-L network of the form*

$$X_{\ell+1} = H^{(\ell)} X_\ell + \mathcal{F}_\ell(X_\ell), \qquad \ell = 1, \dots, L, \tag{3.5}$$

where $H^{(\ell)} \in \mathbb{R}^{n\times n}$ mixes streams and $\mathcal{F}_\ell$ denotes the remaining nonlinear part of the block. Suppose

$$\mathsf{Amax}(\mathcal{F}_\ell(X) - \mathcal{F}_\ell(Y)) \le a_\ell \mathsf{Amax}(X - Y) \tag{3.6}$$

for all relevant X, Y. Then

$$\mathsf{Amax}(X_L - Y_L) \le \prod_{\ell=1}^{L} (\|H^{(\ell)}\|_\infty + a_\ell)\, \mathsf{Amax}(X_0 - Y_0). \tag{3.7}$$

If, in addition, along the realized trajectory the adjoint Jacobian of the nonlinear branch satisfies

$$\mathsf{Amax}(D\mathcal{F}_\ell(X_\ell)^* G) \le b_\ell \mathsf{Amax}(G), \tag{3.8}$$

then the input-gradient amplification obeys

$$\mathsf{Amax}(\nabla_{X_0}\mathcal{L}) \le \prod_{\ell=1}^{L} (\|H^{(\ell)}\|_1 + b_\ell)\, \mathsf{Amax}(\nabla_{X_L}\mathcal{L}). \tag{3.9}$$

In particular, if $H^{(\ell)} \ge 0$ is $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$-doubly stochastic, then

$$\mathsf{Amax}(\nabla_{X_0}\mathcal{L}) \le \prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)} + b_\ell)\, \mathsf{Amax}(\nabla_{X_L}\mathcal{L}). \tag{3.10}$$

Proof. Subtracting two trajectories and using the triangle inequality gives

$$\begin{aligned}\mathsf{Amax}\,(X_{\ell+1}-Y_{\ell+1}) &\le \mathsf{Amax}\,(H^{(\ell)}(X_\ell-Y_\ell)) + \mathsf{Amax}\,(\mathcal{F}_\ell(X_\ell)-\mathcal{F}_\ell(Y_\ell))\\ &\le (\|H^{(\ell)}\|_\infty + a_\ell)\mathsf{Amax}\,(X_\ell-Y_\ell),\end{aligned}$$

which iterates to (3.7). The backward recursion has the form

$$G_\ell = (H^{(\ell)})^\top G_{\ell+1} + D\mathcal{F}_\ell(X_\ell)^* G_{\ell+1}.$$

Using (2.7) and (3.8) gives

$$\mathsf{Amax}\,(G_\ell) \le (\|H^{(\ell)}\|_1 + b_\ell)\mathsf{Amax}\,(G_{\ell+1}),$$

and iteration yields (3.9). The marginal version follows from (3.1). □

The mixer certificate controls the stream-mixing pathway. A full-network certificate additionally requires independent control, measurement, or regularization of the nonlinear-branch constants a_ℓ and b_ℓ in (3.6)–(3.8).

Remark 3.4 (Relation to the detailed mHC Jacobian bound). Proposition 3.3 is architecture-agnostic. Appendix E gives the corresponding row-major Jacobian calculation for the specific mHC layer form (2.1), including the pre/post aggregation vectors and the feature-block Jacobian.

Remark 3.5 (Depth sensitivity and tolerance allocation). The certificate (3.2) is worst-case and therefore depth sensitive. For small errors, $\prod_{\ell=1}^{L}(1+\varepsilon_r^{(\ell)}) = \exp(\sum_\ell \log(1+\varepsilon_r^{(\ell)})) \approx \exp(\sum_\ell \varepsilon_r^{(\ell)})$, and similarly for the backward bound. Thus, to enforce a forward budget $\|T_L\|_\infty \le B_\infty$ it is sufficient to allocate tolerances so that $\sum_{\ell=1}^{L}\varepsilon_r^{(\ell)} \lesssim \log B_\infty$ (and analogously for $\|T_L\|_1$). Subsection 4.7 summarizes a practical allocation rule and its translation into a finite Sinkhorn iteration budget.

The bounds in Proposition 3.1 apply once the marginal errors $\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)}$ are controlled by the mechanism generating $H_{\mathrm{res}}^{(\ell)}$. Section 4 gives explicit finite-step Sinkhorn bounds that map the logit dynamic range and the iteration count $t_{\max}$ to certified marginal tolerances, turning (3.2) into a design-time rule.

Necessity of nonnegativity Proposition 3.1 uses entrywise nonnegativity only through the step (3.1), which turns marginal constraints into induced-norm bounds. A simple example shows that this step fails completely once signs are allowed: even *exact* row and column sums do not prevent arbitrarily large induced norms.

The 2×2 example

$$A_N = \begin{pmatrix} N & 1-N \\ 1-N & N \end{pmatrix}, \qquad N>1, \tag{3.11}$$

has exact row and column sums equal to one, but

$$\|A_N\|_\infty = \|A_N\|_1 = 2N-1. \tag{3.12}$$

Since A_N is symmetric, also $\|A_N\|_2 = 2N-1$. Thus exact marginals alone do not control induced norms once signs are allowed; embedding this block into a larger identity matrix gives the same conclusion for every $n \ge 2$.

Remark 3.6 (Implication for mHC certificates). This shows that marginal constraints alone provide no worst-case protection against amplification when signs are allowed. By contrast, for $P \in \mathcal{B}_n$ (entrywise nonnegative and exactly doubly stochastic) one has

$$\|P\|_\infty = \|P\|_1 = 1 \quad \text{and hence} \quad \|P\|_2 \le 1, \tag{3.13}$$

so worst-case amplification through the mixer pathway is ruled out under exact marginals. This is why enforcing positivity (via exponentiating logits) is not merely a convenience for Sinkhorn convergence, but a structural requirement for worst-case certificates based on marginal errors.

Beyond mixer-path certificates. The mixer-path estimates are the primitive certificates. Proposition 3.3 shows how to combine them with Amax Lipschitz bounds for the nonlinear residual branch, and Appendix E gives the corresponding row-major Jacobian formula for the specific mHC block.

4 Finite-step Sinkhorn: bounds and calibration

Proposition 3.1 bounds the depth-wise mixer gains in terms of per-layer marginal errors. Here we relate those errors to the finite-step Sinkhorn iterations used in the mHC-motivated mixer parameterization considered here, giving a conservative rule for choosing $(\kappa, t_{\max})$ to meet a prescribed depth-L budget.

Standing assumptions for the finite-step certificate

The finite-step Sinkhorn certificates below are proved under the following assumptions.

1. **Strict positivity and dense support.** The pre-scaling matrix satisfies $S^{(0)} \in \mathbb{R}^{n\times n}_{>0}$. Masked or sparse support requires a separate support-dependent analysis.

2. **Dynamic range control.** The entrywise dynamic range is bounded:

$$\kappa := \frac{\max_{i,j} S^{(0)}_{ij}}{\min_{i,j} S^{(0)}_{ij}} < \infty, \tag{4.1}$$

   enforced in applications by exponentiating clipped logits, for example $S^{(0)}_{ij} = \exp(\beta C_{ij})$ with $C_{ij} \in [-R, R]$.

3. **Column-then-row Sinkhorn order.** A *full step* means `column-then-row` normalization. We cap computation by a budget $t_{\max}$. The analysis allows any chosen number of full steps $t^{(\ell)} \in \{1, \dots, t_{\max}\}$ (possibly layer-dependent), producing the layer mixer

$$H^{(\ell)}_{\text{res}} = S^{(2t^{(\ell)})}. \tag{4.2}$$

   The fixed-step variant uses $t^{(\ell)} = t_{\max}$, while the early-stopping variant uses $t^{(\ell)} = t^{(\ell)}_{\text{stop}} \le t_{\max}$ (Subsection 4.2).

4. **Exact arithmetic versus implemented arithmetic.** The theorems use the exact normalization maps T_c and T_r. If an implementation uses denominator stabilizers or floors, then certificates must be evaluated using the actually realized row and column defects of the implemented matrix.

5. **Mixer-path scope.** The resulting norm bounds certify the stream-mixing product $T_L = H_{\text{res}}^{(L)} \cdots H_{\text{res}}^{(1)}$. A full-network certificate additionally requires separate control of the nonlinear residual branch.

Recall that the idealized `column-then-row` full-step order ends with a row normalization. Hence the mathematical output after each full step has exact row sums. In code using denominator stabilizers, we measure the residual row defect and include it in the reported marginal diagnostics.

4.1 Alternating normalization and marginal errors

Given logits $\widetilde{H_{\text{res}}}^{(\ell)} \in \mathbb{R}^{n\times n}$, the mixer parameterization forms a positive matrix

$$S^{(0)} := \exp(\beta\, \widetilde{H_{\text{res}}}^{(\ell)}) \qquad \text{(entrywise)}, \tag{4.3}$$

where $\beta > 0$ is an optional temperature. We then apply alternating column and row normalizations:

$$S^{(s+1)} := \begin{cases} T_c(S^{(s)}) & s \text{ even}, \\ T_r(S^{(s)}) & s \text{ odd}, \end{cases} \qquad s = 0, 1, 2, \ldots \tag{4.4}$$

where T_c divides each column by its column sum and T_r divides each row by its row sum. After a column normalization we have exact column sums, and after a row normalization we have exact row sums:

$$\mathbf{1}^\top S^{(s)} = \mathbf{1}^\top \ \ (s \text{ odd}), \qquad S^{(s)}\mathbf{1} = \mathbf{1} \ \ (s \text{ even},\ s \geq 2). \tag{4.5}$$

Accordingly, the nontrivial marginal errors occur on alternating half-steps. We define

$$\varepsilon_r^{(s)} := \left\| S^{(s)}\mathbf{1} - \mathbf{1} \right\|_{\infty,\text{vec}}, \qquad \varepsilon_c^{(s)} := \left\| \mathbf{1}^\top S^{(s)} - \mathbf{1}^\top \right\|_{\infty,\text{vec}}. \tag{4.6}$$

If we run t *full* Sinkhorn steps in the `column-then-row` order, this corresponds to $2t$ alternating normalizations and the output

$$H_{\text{res}}^{(\ell)} := S^{(2t)}. \tag{4.7}$$

We use either (i) a fixed-step choice $t = t_{\max}$, or (ii) an early-stopped choice $t = t_{\text{stop}}^{(\ell)} \leq t_{\max}$ where $t_{\text{stop}}^{(\ell)}$ is the smallest full-step count achieving the target column-marginal tolerance (Subsection 4.2).

Layer-level errors used in Proposition 3.1. For a layer output $H_{\text{res}}^{(\ell)} = S^{(2t)}$ (with t given by $t^{(\ell)}$ in (4.2)) define $\varepsilon_r^{(\ell)} := \|H_{\text{res}}^{(\ell)}\mathbf{1} - \mathbf{1}\|_{\infty,\text{vec}}$ and $\varepsilon_c^{(\ell)} := \|\mathbf{1}^\top H_{\text{res}}^{(\ell)} - \mathbf{1}^\top\|_{\infty,\text{vec}}$.

Lemma 4.1 (Ideal output row sums and induced ℓ_∞ control)**.** *For the exact arithmetic normalization maps T_c and T_r, under the* `column-then-row` *order and for any $t \geq 1$, the output $H_{\mathrm{res}}^{(\ell)} := S^{(2t)}$ satisfies*

$$H_{\mathrm{res}}^{(\ell)} \mathbf{1} = \mathbf{1} \tag{4.8}$$

If moreover $H_{\mathrm{res}}^{(\ell)} \geq 0$, then

$$\|H_{\mathrm{res}}^{(\ell)}\|_\infty = 1, \tag{4.9}$$

so the only nontrivial marginal error for the backward certificate is the output column error $\varepsilon_c^{(\ell)} = \|\mathbf{1}^\top H_{\mathrm{res}}^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}$, which controls $\|H_{\mathrm{res}}^{(\ell)}\|_1$.

Proof. Each full step ends with a row normalization, hence (4.8). If $H_{\mathrm{res}}^{(\ell)} \geq 0$, then $\|H_{\mathrm{res}}^{(\ell)}\|_\infty = \max_i \sum_j (H_{\mathrm{res}}^{(\ell)})_{ij} = 1$. □

Under the `column-then-row` schedule, the forward mixer-path Amax gain is certified layerwise automatically, and the practically active design problem is the backward one: controlling $\|H_{\mathrm{res}}^{(\ell)}\|_1$ via the output column-marginal error $\varepsilon_c^{(\ell)} = \|\mathbf{1}^\top H_{\mathrm{res}}^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}$.

4.2 Early stopping to a target column tolerance

At each layer, we run full `column-then-row` Sinkhorn steps and measure the output column defect

$$\varepsilon_c^{(\ell)}(t) := \|\mathbf{1}^\top S^{(2t)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}. \tag{4.10}$$

The layer stops at the first full-step count

$$t_{\mathrm{stop}}^{(\ell)} = \min\{1 \leq t \leq t_{\max} : \varepsilon_c^{(\ell)}(t) \leq \bar{\varepsilon}_\ell\}, \tag{4.11}$$

if such a step exists, and otherwise returns the $t_{\max}$ output with a failed certificate for that layer. Appendix A gives the corresponding implementation pseudocode. The important convention is that $\varepsilon_c^{(\ell)}(t)$ is measured after a full column-then-row step, namely at $S^{(2t)}$.

Remark 4.2 (Additive denominator stabilizers)*.* The theoretical statements below use the exact normalization maps, i.e. division by the current row and column sums. If an implementation divides by $s + \epsilon$ rather than by s, then the corresponding marginal is no longer exactly normalized. For example, after a row normalization with row sum $s_i > 0$, the new row sum is $s_i/(s_i + \epsilon)$, and the row defect is $\epsilon/(s_i + \epsilon)$. In our experiments $\epsilon = 10^{-12}$, so this effect is numerically negligible. Strictly speaking, however, certificates for the implemented matrix should use the actually measured row and column defects; all implementation-facing diagnostics below are interpreted in this measured sense.

4.3 Dynamic range and Hilbert-metric control

Assume $S^{(0)}$ has bounded dynamic range: there exist $0 < m \leq M$ such that $m \leq S_{ij}^{(0)} \leq M$ for all i, j, and define $\kappa := M/m$. If $S^{(0)} = \exp(\beta C)$, then

$$\kappa = \exp\big(\beta(\max_{i,j} C_{ij} - \min_{i,j} C_{ij})\big), \tag{4.12}$$

so clipping $C_{ij} \in [-R, R]$ implies $\kappa \le \exp(2\beta R)$.

Recall that for $u, v \in \mathbb{R}^n_{>0}$, the Hilbert projective metric is

$$d_H(u, v) := \log\Big(\max_i \frac{u_i}{v_i}\Big) - \log\Big(\min_i \frac{u_i}{v_i}\Big). \tag{4.13}$$

Lemma 4.3 (Explicit initial spread)**.** *If $m \le S^{(0)}_{ij} \le M$ and $u := S^{(0)}\mathbf{1}$, then $d_H(u, \mathbf{1}) \le \log \kappa$. Similarly, for $v := \mathbf{1}^\top S^{(0)}$ we have $d_H(v^\top, \mathbf{1}) \le \log \kappa$.*

Proof. Each component $u_i = \sum_j S^{(0)}_{ij}$ lies in $[nm, nM]$, so $\max_i u_i / \min_i u_i \le \kappa$ and $d_H(u, \mathbf{1}) = \log(\max_i u_i / \min_i u_i) \le \log \kappa$. The column-sum statement is identical. □

Lemma 4.4 (Hilbert-to-marginal conversion)**.** *Let $u \in \mathbb{R}^n_{>0}$ satisfy $\frac{1}{n}\sum_{i=1}^n u_i = 1$ and set $\delta := d_H(u, \mathbf{1}) = \log(\max_i u_i / \min_i u_i)$. Then*

$$\|u - \mathbf{1}\|_{\infty,\mathrm{vec}} \le \Psi_n(\delta) := \frac{(n-1)(e^\delta - 1)}{(n-1) + e^\delta}. \tag{4.14}$$

Moreover, for any $\varepsilon \in [0, n-1)$,

$$\Psi_n(\delta) \le \varepsilon \quad \Longleftrightarrow \quad \delta \le \Psi_n^{-1}(\varepsilon) := \log\Big(\frac{(n-1)(1+\varepsilon)}{(n-1) - \varepsilon}\Big). \tag{4.15}$$

Proof. See Appendix B. □

Lemma 4.5 (Centered-log L^2 conversion to marginal error)**.** *Let $u \in \mathbb{R}^n_{>0}$ satisfy $n^{-1}\sum_i u_i = 1$ and put*

$$z := \Pi \log u, \qquad \eta := \|z\|_2.$$

Then

$$\|u - \mathbf{1}\|_{\infty,\mathrm{vec}} \le \Gamma_n(\eta), \tag{4.16}$$

where

$$\Gamma_n(\eta) := \frac{n \exp(a)}{\exp(a) + (n-1)\exp(-a/(n-1))} - 1, \qquad a := \sqrt{\frac{n-1}{n}}\,\eta. \tag{4.17}$$

The bound is sharp for the upper tail: equality is attained when one centered log-coordinate equals a and the remaining $n-1$ centered log-coordinates equal $-a/(n-1)$. For small η,

$$\Gamma_n(\eta) = \sqrt{\frac{n-1}{n}}\,\eta + O(\eta^2). \tag{4.18}$$

Proof. See Appendix B. □

4.4 A finite-step sufficient rule

Classical Birkhoff–Bushell contraction theory for positive maps implies geometric contraction in the Hilbert metric with a rate controlled by the dynamic range (Birkhoff, 1957; Bushell, 1973). Specializing to the alternating normalizations above yields the following conservative but explicit design bound.

Half-step indexing. For clarity we record the indexing convention used throughout the finite-step analysis. Starting from $S^{(0)} = A$, the alternating normalizations are

$$S^{(1)} = T_c(A), \qquad S^{(2)} = T_r T_c(A), \qquad S^{(3)} = T_c T_r T_c(A), \qquad \dots. \tag{4.19}$$

Thus $S^{(2t)}$ is the output after t full column-then-row steps. In exact arithmetic, $S^{(2t)}$ has exact row sums, whereas its column sums are generally only approximate. This is the origin of the one-half-step loss in the output column-error estimate: the column error at $S^{(2t)}$ is controlled at rate τ^{2t-1}, not τ^{2t}.

Lemma 4.6 (Birkhoff–Bushell contraction for positive matrices)**.** *Let $A \in \mathbb{R}^{n\times n}_{>0}$. Define its projective diameter by*

$$\Delta(A) := \log \max_{i,k,j,\ell} \frac{A_{ij}A_{k\ell}}{A_{i\ell}A_{kj}}, \tag{4.20}$$

and set

$$\tau(A) := \tanh\left(\frac{\Delta(A)}{4}\right). \tag{4.21}$$

Then, for all $x, y \in \mathbb{R}^n_{>0}$,

$$d_H(Ax, Ay) \le \tau(A) d_H(x, y), \qquad d_H(A^\top x, A^\top y) \le \tau(A) d_H(x, y). \tag{4.22}$$

Moreover, componentwise inversion is an isometry for the Hilbert metric:

$$d_H(x^{-1}, y^{-1}) = d_H(x, y), \tag{4.23}$$

where x^{-1} denotes the componentwise reciprocal.

Proof. The first estimate is the classical Birkhoff–Bushell contraction theorem for positive linear maps in Hilbert's projective metric. Since A and $A^\top$ have the same projective diameter, the same contraction coefficient applies to $A^\top$. Finally,

$$d_H(x^{-1}, y^{-1}) = \log \frac{\max_i y_i/x_i}{\min_i y_i/x_i} = \log \frac{\max_i x_i/y_i}{\min_i x_i/y_i} = d_H(x, y).$$

□

Theorem 4.7 (Finite-step marginal-error bound from projective contraction)**.** *Let $A := S^{(0)} \in \mathbb{R}^{n\times n}_{>0}$, $n \ge 2$, and let $S^{(s)}$ be the alternating normalization iterates from Subsection 4.1. Define the projective diameter*

$$\Delta(A) := \log \max_{i,k,j,\ell} \frac{A_{ij}A_{k\ell}}{A_{i\ell}A_{kj}}, \tag{4.24}$$

and set

$$\tau(A) := \tanh\left(\frac{\Delta(A)}{4}\right). \tag{4.25}$$

Let

$$P^{(1)} := T_c(A), \qquad \delta_0 := d_H(P^{(1)}\mathbf{1}, \mathbf{1}). \tag{4.26}$$

Then, for every integer $t \geq 0$,

$$\varepsilon_r^{(2t+1)} := \|S^{(2t+1)}\mathbf{1} - \mathbf{1}\|_{\infty,\mathrm{vec}} \leq \Psi_n\Big(\delta_0 \tau(A)^{2t}\Big). \tag{4.27}$$

Moreover, for every integer $t \geq 1$,

$$\varepsilon_c^{(2t)} := \|\mathbf{1}^\top S^{(2t)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}} \leq \Psi_n\Big(\delta_0 \tau(A)^{2t-1}\Big). \tag{4.28}$$

In particular, if A has entrywise dynamic range

$$\kappa = \frac{\max_{i,j} A_{ij}}{\min_{i,j} A_{ij}},$$

then

$$\tau(A) \leq q := \frac{\kappa - 1}{\kappa + 1}, \qquad \delta_0 \leq \log \kappa,$$

and hence

$$\varepsilon_r^{(2t+1)} \leq \Psi_n\Big((\log \kappa) q^{2t}\Big), \qquad t \geq 0, \tag{4.29}$$

while

$$\varepsilon_c^{(2t)} \leq \Psi_n\Big((\log \kappa) q^{2t-1}\Big), \qquad t \geq 1. \tag{4.30}$$

Consequently, to guarantee $\varepsilon_c^{(2t_{\max})} \leq \bar{\varepsilon}$, it is sufficient that

$$t_{\max} \geq \frac{1}{2}\left[1 + \frac{\log\big((\log \kappa)/\Psi_n^{-1}(\bar{\varepsilon})\big)}{\log(1/q)}\right], \tag{4.31}$$

with the right-hand side rounded up and with $t_{\max} \geq 1$ unless A is already normalized. If $\kappa = 1$, then A is a constant matrix and one full step gives an exactly doubly stochastic output.

Proof. Let $r^{(t)} \in \mathbb{R}^n_{>0}$ be the accumulated row-scaling vector at the beginning of the t-th full column-then-row step, with $r^{(0)} = \mathbf{1}$, modulo an irrelevant scalar factor. After the column normalization in full step t,

$$S^{(2t+1)} = \mathrm{diag}(r^{(t)}) A \,\mathrm{diag}\Big((A^\top r^{(t)})^{-1}\Big).$$

Its row-sum vector is therefore

$$u_t := S^{(2t+1)}\mathbf{1} = r^{(t)} \odot A(A^\top r^{(t)})^{-1}.$$

The following row normalization divides by this row-sum vector, hence

$$r^{(t+1)} = r^{(t)} \oslash u_t = \Big[A(A^\top r^{(t)})^{-1}\Big]^{-1} =: \Phi(r^{(t)}).$$

We first prove that Φ contracts in Hilbert distance with coefficient $\tau(A)^2$. For $r, s \in \mathbb{R}^n_{>0}$, Lemma 4.6 gives

$$\begin{aligned} d_H(A^\top r, A^\top s) &\leq \tau(A) d_H(r, s), \\ d_H((A^\top r)^{-1}, (A^\top s)^{-1}) &= d_H(A^\top r, A^\top s), \\ d_H(A(A^\top r)^{-1}, A(A^\top s)^{-1}) &\leq \tau(A) d_H((A^\top r)^{-1}, (A^\top s)^{-1}). \end{aligned}$$

Applying the inversion isometry once more yields

$$d_H(\Phi(r), \Phi(s)) \le \tau(A)^2 d_H(r, s).$$

Consequently,

$$d_H(r^{(t+1)}, r^{(t)}) \le \tau(A)^{2t} d_H(r^{(1)}, r^{(0)}).$$

Since $u_t = r^{(t)} \oslash r^{(t+1)}$,

$$d_H(u_t, \mathbf{1}) = d_H(r^{(t)}, r^{(t+1)}) \le \tau(A)^{2t}\delta_0, \qquad \delta_0 = d_H(u_0, \mathbf{1}) = d_H(P^{(1)}\mathbf{1}, \mathbf{1}).$$

After a column normalization the total mass is n, so $n^{-1}\sum_i (u_t)_i = 1$. Lemma 4.4 gives (4.27).

It remains to control the column sums at the full-step output $S^{(2t)}$. For $t \ge 1$, write

$$v_t := (\mathbf{1}^\top S^{(2t)})^\top.$$

Using the row-scaled representation before and after the preceding row normalization gives

$$v_t = (A^\top r^{(t)}) \oslash (A^\top r^{(t-1)}).$$

Therefore

$$d_H(v_t, \mathbf{1}) = d_H(A^\top r^{(t)}, A^\top r^{(t-1)}) \le \tau(A) d_H(r^{(t)}, r^{(t-1)}) \le \delta_0 \tau(A)^{2t-1}.$$

Since $S^{(2t)}$ has total mass n, Lemma 4.4 gives (4.28).

Finally, if $m \le A_{ij} \le M$ and $\kappa = M/m$, then $\Delta(A) \le 2\log\kappa$, hence

$$\tau(A) \le \tanh\left(\frac{\log\kappa}{2}\right) = \frac{\kappa - 1}{\kappa + 1}.$$

Also, if $P^{(1)} = T_c(A)$, then for any rows i, k,

$$\frac{(P^{(1)}\mathbf{1})_i}{(P^{(1)}\mathbf{1})_k} = \frac{\sum_j P^{(1)}_{ij}}{\sum_j P^{(1)}_{kj}} \le \max_j \frac{P^{(1)}_{ij}}{P^{(1)}_{kj}} = \max_j \frac{A_{ij}}{A_{kj}} \le \kappa.$$

Thus $\delta_0 \le \log\kappa$. The step-count condition follows by solving

$$(\log\kappa) q^{2t_{\max}-1} \le \Psi_n^{-1}(\bar{\varepsilon}).$$

□

Corollary 4.8 (Output bound for the column-then-row order)**.** *Under the ideal exact* `column-then-row` *schedule with output $H^{(\ell)}_{\mathrm{res}} = S^{(2t_{\max})}$, the output satisfies $H^{(\ell)}_{\mathrm{res}}\mathbf{1} = \mathbf{1}$ and hence $\|H^{(\ell)}_{\mathrm{res}}\|_\infty = 1$ for $H^{(\ell)}_{\mathrm{res}} \ge 0$. In implemented normalizations with denominator stabilizers, this exact row-sum statement is replaced by the measured row defect described in Remark 4.2. Moreover, in the ideal exact schedule the output column-marginal error obeys*

$$\varepsilon_c^{(\ell)} = \left\|\mathbf{1}^\top H^{(\ell)}_{\mathrm{res}} - \mathbf{1}^\top\right\|_{\infty,\mathrm{vec}} \le \Psi_n\left((\log\kappa) q^{2t_{\max}-1}\right), \qquad q = \frac{\kappa - 1}{\kappa + 1}. \tag{4.32}$$

Thus certifying the backward mixing-path gain reduces to choosing $t_{\max}$ so that the right-hand side of (4.32) is at most the desired per-layer tolerance.

Remark 4.9 (Conservatism of the step-count bound). Theorem 4.7 is a worst-case certificate: it guarantees a target marginal tolerance for any strictly positive input matrix with dynamic range at most κ. The matrix-adaptive version (4.28), based on $\Delta(A)$ and δ_0, is often sharper than the κ-only bound. In practice, both should be viewed as safe certificates rather than numerical predictions; online early stopping based on the measured marginal error is usually much less pessimistic.

4.5 Matrix-specific sharp prediction and local certificates

The dynamic-range estimate in Theorem 4.7 is a uniform worst-case certificate. It is safe but not numerically sharp for typical random logits. The sharper object is the nonlinear Sinkhorn map in logarithmic projective coordinates. Its derivative at the Sinkhorn limit gives the asymptotic rate $\sigma_2(P)^2$, while its derivative on a small ball gives a non-asymptotic, matrix-specific certificate that is much closer to the measured curve.

Let

$$P = \operatorname{diag}(r_*)A\operatorname{diag}(c_*) \in \mathcal{B}_n$$

be the limiting doubly stochastic scaling of $A \in \mathbb{R}^{n\times n}_{>0}$. We work on $\mathbf{1}^\perp$ and define

$$\mathcal{F}(x) := \Pi\left[-\log\left(P\left((P^\top e^x)^{-1}\right)\right)\right], \qquad \mathcal{L}(x) := \Pi\log(P^\top e^x), \tag{4.33}$$

where all exponentials, logarithms, and inverses are componentwise. If

$$x_t := \Pi\log\frac{r^{(t)}}{r_*},$$

then

$$x_{t+1} = \mathcal{F}(x_t), \tag{4.34}$$

and the centered logarithmic output column error after $t \geq 1$ full steps is

$$z_t := \Pi\log v_t = \mathcal{L}(x_t) - \mathcal{L}(x_{t-1}), \qquad v_t := (\mathbf{1}^\top S^{(2t)})^\top. \tag{4.35}$$

Theorem 4.10 (Sharp local spectral rate)**.** *Let $A \in \mathbb{R}^{n\times n}_{>0}$ and let $P = \operatorname{diag}(r_*)A\operatorname{diag}(c_*) \in \mathcal{B}_n$ be its Sinkhorn limit. Set*

$$K := PP^\top, \qquad \rho_* := \|K\Pi\|_{2\to 2}.$$

Then

$$\rho_* = \sigma_2(P)^2. \tag{4.36}$$

Moreover,

$$D\mathcal{F}(0) = K|_{\mathbf{1}^\perp}, \tag{4.37}$$

so, locally,

$$x_{t+1} = Kx_t + O(\|x_t\|^2). \tag{4.38}$$

Consequently, for every $\eta > 0$ there is a neighborhood of zero such that

$$\|x_t\|_2 \leq C_\eta(\rho_* + \eta)^t\|x_0\|_2. \tag{4.39}$$

Finally,

$$z_t = \Pi P^\top (K - I) K^{t-1} x_0 + O\Big((\rho_* + \eta)^t \|x_0\|_2^2\Big) \tag{4.40}$$

locally. In the generic case where x_0 has a nonzero projection onto the leading nontrivial eigenspace of K, the small-error ratio is governed by

$$\frac{\|z_{t+1}\|}{\|z_t\|} \longrightarrow \rho_* = \sigma_2(P)^2$$

up to the usual modification when the leading nontrivial eigenvalue has multiplicity.

Proof. The recurrence (4.34) follows from the identities

$$A^\top r^{(t)} = c_*^{-1} \odot P^\top e^{x_t}, \qquad (A^\top r^{(t)})^{-1} = c_* \odot (P^\top e^{x_t})^{-1},$$

where the irrelevant scalar in e^{x_t} is removed by Π. The row-scaling update gives exactly

$$e^{x_{t+1}} \sim \Big[P\Big((P^\top e^{x_t})^{-1}\Big)\Big]^{-1},$$

and hence (4.34). Since $P^\top \mathbf{1} = \mathbf{1}$,

$$P^\top e^x = \mathbf{1} + P^\top x + O(\|x\|^2),$$

so

$$(P^\top e^x)^{-1} = \mathbf{1} - P^\top x + O(\|x\|^2).$$

Applying P and taking $-\log$ gives

$$\mathcal{F}(x) = \Pi P P^\top x + O(\|x\|^2).$$

On $\mathbf{1}^\perp$, $K = PP^\top$ is symmetric and has operator norm $\sigma_2(P)^2$, proving (4.36)–(4.38). The perturbative estimate (4.39) is the standard consequence of a C^2 map whose derivative at the fixed point has spectral radius ρ_*. Formula (4.35) gives

$$z_t = \mathcal{L}(x_t) - \mathcal{L}(x_{t-1}).$$

Since $D\mathcal{L}(0) = \Pi P^\top$ and $x_t = K^t x_0 + O((\rho_* + \eta)^t \|x_0\|^2)$ locally, (4.40) follows. □

The preceding theorem is asymptotic. The next result is the sharper finite-step version that should be used when one wants a certificate close to the numerical curve.

Theorem 4.11 (Non-asymptotic local spectral envelope)**.** *For $x \in \mathbf{1}^\perp$, define*

$$g(x) := P^\top e^x, \qquad a(x) := P(g(x)^{-1}),$$

and set

$$J_F(x) := \Pi \operatorname{diag}(a(x))^{-1} P \operatorname{diag}(g(x))^{-2} P^\top \operatorname{diag}(e^x) \Pi, \tag{4.41}$$

$$J_L(x) := \Pi \operatorname{diag}(g(x))^{-1} P^\top \operatorname{diag}(e^x) \Pi. \tag{4.42}$$

For $R > 0$, put

$$\rho_R := \sup_{\substack{x \in \mathbf{1}^\perp \\ \|x\|_2 \le R}} \|J_F(x)\|_{2\to 2}, \qquad \lambda_R := \sup_{\substack{x \in \mathbf{1}^\perp \\ \|x\|_2 \le R}} \|J_L(x)\|_{2\to 2}. \tag{4.43}$$

The quantities ρ_R and λ_R are exact local derivative envelopes. Any computable upper bounds $\widehat{\rho}_R \ge \rho_R$ and $\widehat{\lambda}_R \ge \lambda_R$ may be substituted in the conclusion, provided $\widehat{\rho}_R < 1$. Assume $\rho_R < 1$ and $\|x_0\|_2 \le R$. Then the full Sinkhorn orbit remains in the ball $\|x\|_2 \le R$ and, for every $t \ge 1$,

$$\|z_t\|_2 \le \lambda_R \rho_R^{t-1} \|x_1 - x_0\|_2. \tag{4.44}$$

Consequently, the actual column-marginal error satisfies the following local finite-step bound

$$\varepsilon_c^{(2t)} \le \Gamma_n\Big(\lambda_R \rho_R^{t-1} \|x_1 - x_0\|_2\Big), \tag{4.45}$$

where Γ_n is defined in Lemma 4.5. The same conclusion holds with ρ_R, λ_R replaced by any certified upper bounds $\widehat{\rho}_R, \widehat{\lambda}_R$ satisfying $\widehat{\rho}_R < 1$.

Proof. The displayed formulae for J_F and J_L are obtained by differentiating (4.33). Because $\mathcal{F}(0) = 0$, the mean-value theorem gives, for any $x \in \mathbf{1}^\perp$ with $\|x\|_2 \le R$,

$$\|\mathcal{F}(x)\|_2 \le \rho_R \|x\|_2.$$

Thus the ball is invariant whenever $\rho_R < 1$. Applying the same mean-value estimate to the differences gives

$$\|x_{s+1} - x_s\|_2 \le \rho_R \|x_s - x_{s-1}\|_2, \qquad s \ge 1,$$

and therefore

$$\|x_t - x_{t-1}\|_2 \le \rho_R^{t-1} \|x_1 - x_0\|_2.$$

Since $z_t = \mathcal{L}(x_t) - \mathcal{L}(x_{t-1})$, another mean-value estimate gives (4.44). Lemma 4.5 converts the centered logarithmic error into the ordinary column-sum error. □

Corollary 4.12 (Orbit-wise derivative certificate)**.** *Let $x_0, x_1, \dots, x_t$ be the actual logarithmic Sinkhorn orbit and define*

$$\rho_s := \sup_{0 \le \theta \le 1} \big\|J_F\big((1-\theta)x_{s-1} + \theta x_s\big)\big\|_{2\to 2}, \qquad s \ge 1,$$

$$\lambda_t^{\mathrm{seg}} := \sup_{0 \le \theta \le 1} \big\|J_L\big((1-\theta)x_{t-1} + \theta x_t\big)\big\|_{2\to 2}.$$

Then

$$\|z_t\|_2 \le \lambda_t^{\mathrm{seg}} \left(\prod_{s=1}^{t-1} \rho_s\right) \|x_1 - x_0\|_2, \tag{4.46}$$

and hence

$$\varepsilon_c^{(2t)} \le \Gamma_n\left(\lambda_t^{\mathrm{seg}} \left(\prod_{s=1}^{t-1} \rho_s\right) \|x_1 - x_0\|_2\right). \tag{4.47}$$

This is an a posteriori certificate along the realized orbit; it replaces the global dynamic-range rate by the actual derivative norms encountered by the matrix.

Proof. Apply the mean-value theorem on each segment joining x_{s-1} to x_s and then on the segment joining x_{t-1} to x_t for $\mathcal{L}$. The final conversion is Lemma 4.5. □

4.6 Certificates versus predictors

The estimates in this section play different roles. We separate them explicitly because only some of them are deterministic certificates.

Quantity	Certificate?	Scope	Use
κ-only Hilbert bound	Yes	Uniform over positive matrices with dynamic range at most κ	Design-time worst-case guarantee
$\Delta(A), \delta_0$ Hilbert bound	Yes	Fixed positive input matrix	Sharper a priori certificate
ρ_R, λ_R derivative envelope	Yes, conditional	Verified ball around the Sinkhorn fixed point	Local non-asymptotic certificate
$\rho_* = \sigma_2(P)^2$	Asymptotic	Near the Sinkhorn limit	Sharp local slope prediction
First- and second-order expansions	No	Small-error perturbative regime	Numerical predictors for observed finite-step decay

Thus the worst-case and derivative-envelope estimates may be used to certify a stability budget. The first- and second-order formulas are not stability certificates; they are used only to explain the observed convergence curve and the gap between worst-case and matrix-specific behavior.

First-order local spectral predictor. The leading predictor used for plots is

$$\ell_t^{[1]} := \Pi P^\top (K - I) K^{t-1} x_0, \qquad K = PP^\top, \tag{4.48}$$

followed by exponentiation and recentering:

$$\widehat{v}_t^{[1]} := \frac{n \exp(\ell_t^{[1]})}{\sum_{j=1}^n \exp((\ell_t^{[1]})_j)}, \qquad \widehat{\varepsilon}_{c,1}^{(2t)} := \|\widehat{v}_t^{[1]} - \mathbf{1}\|_{\infty,\mathrm{vec}}. \tag{4.49}$$

For large t this reduces to the scalar law

$$\widehat{\varepsilon}_{c,1}^{(2t)} \approx C_{\mathrm{lin}} \rho_*^{t-1}, \qquad C_{\mathrm{lin}} := \|\Pi P^\top (K - I) x_0\|_{\infty,\mathrm{vec}}.$$

Second-order predictor. The following expansion is not used in any deterministic stability guarantee. It is a small-error predictor for the observed finite-step column-marginal decay of a fixed matrix. Its purpose is to explain numerical behavior near the Sinkhorn limit, not to replace the global or local certificates above.

The linear predictor is sharp asymptotically but can miss the first few Sinkhorn steps. In the diagnostics we therefore also use a quadratic finite-step predictor, obtained by expanding the logarithmic Sinkhorn map and the output column-error map to second order. The formulas for $\widehat{\varepsilon}_{c,2}^{(2t)}$ are given in Appendix C. This predictor is used only for curve interpretation; the certified stability budgets above do not depend on it.

Remark 4.13 (Hierarchy of estimates). The estimates now have four distinct roles. The κ-only bound is a uniform design-time guarantee. The matrix-adaptive Hilbert bound uses $\Delta(A)$ and δ_0 and is a sharper certificate for a fixed input matrix. The local envelope (4.45) is a non-asymptotic certificate on a neighborhood of the Sinkhorn limit and uses the actual derivative norms of the matrix. The first- and second-order predictors (4.49) and (C.6) are not worst-case certificates; they are sharp local numerical predictions.

Remark 4.14 (Stability–expressivity trade-off via logit range). Smaller dynamic range κ (e.g., via clipping or smaller β) improves the contraction factor q and reduces the step budget required to reach a target marginal tolerance. At the same time, it restricts expressivity: as $\kappa \downarrow 1$, the feasible matrices concentrate near the barycenter $J = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, approaching uniform averaging rather than selective routing. Thus κ (or β) trades off stability against routing selectivity.

4.7 Mixer-path calibration from Amax budgets

Combining the depth-wise product certificate (Proposition 3.1) with the finite-step Sinkhorn bound (Theorem 4.7) gives a direct calibration rule from target Amax budgets to implementable parameters (logit dynamic range and step count).

Theorem 4.15 (Mixer-path calibration rule: $(\kappa, t_{\max}) \Rightarrow \|T_L\|_1$ budget)**.** *Assume each mixer $H_{\mathrm{res}}^{(\ell)}$ is entrywise nonnegative and produced by exponentiating logits with dynamic range at most κ, followed by $t_{\max} \geq 1$ ideal exact full Sinkhorn steps in the* `column-then-row` *order. Suppose the desired per-layer column-marginal tolerance is ε_c. If*

$$\Psi_n\Big((\log \kappa) q^{2t_{\max}-1}\Big) \leq \varepsilon_c, \qquad q = \frac{\kappa - 1}{\kappa + 1}, \tag{4.50}$$

then every output mixer satisfies $\varepsilon_c^{(\ell)} \leq \varepsilon_c$ and the backward Amax gain of the depth-L mixer product T_L satisfies

$$\|T_L\|_1 \leq (1 + \varepsilon_c)^L. \tag{4.51}$$

In particular, to enforce a backward budget $\|T_L\|_1 \leq B_1$ it is sufficient to choose

$$\varepsilon_c \leq B_1^{1/L} - 1 \;\approx\; \frac{\log B_1}{L} \qquad (\textit{for small } \varepsilon_c), \tag{4.52}$$

and then choose $t_{\max}$ satisfying (4.50). Equivalently, it is sufficient that

$$t_{\max} \geq \frac{1}{2}\left[1 + \frac{\log\big((\log \kappa)/\Psi_n^{-1}(\varepsilon_c)\big)}{\log(1/q)}\right], \tag{4.53}$$

rounded up, with $t_{\max} \geq 1$ unless the matrix is already normalized.

Proof. The corrected column-marginal estimate (4.30) gives $\varepsilon_c^{(\ell)} \leq \varepsilon_c$ under (4.50). Proposition 3.1 then gives

$$\|T_L\|_1 \leq \prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}) \leq (1 + \varepsilon_c)^L.$$

The tolerance choice (4.52) is exactly the condition $(1 + \varepsilon_c)^L \leq B_1$. Solving (4.50) for $t_{\max}$ gives (4.53). □

4.8 Adaptive global-budget early stopping

The uniform rule $\varepsilon_c \le B_1^{1/L} - 1$ is simple, but it is not step-efficient: some layers reach a much smaller error than required, while other layers need more iterations. A practical alternative is to allocate the logarithmic product budget online.

Lemma 4.16 (Adaptive log-budget early stopping)**.** *Fix a target backward budget $B_1 \ge 1$ and set $b_1 = \log B_1$. Suppose the layers are processed in order. Define $s_0 = 0$, and before layer ℓ set*

$$\bar{\varepsilon}_\ell := \exp\left(\frac{b_1 - s_{\ell-1}}{L - \ell + 1}\right) - 1. \tag{4.54}$$

Run finite-step Sinkhorn at layer ℓ until

$$\varepsilon_c^{(\ell)} \le \bar{\varepsilon}_\ell, \tag{4.55}$$

and then update

$$s_\ell = s_{\ell-1} + \log(1 + \varepsilon_c^{(\ell)}). \tag{4.56}$$

If every layer reaches its assigned tolerance before the step cap, then

$$\sum_{\ell=1}^{L} \log(1 + \varepsilon_c^{(\ell)}) \le b_1, \qquad \|T_L\|_1 \le B_1. \tag{4.57}$$

Proof. Let $r_{\ell-1} := b_1 - s_{\ell-1}$ be the remaining log-budget. By construction,

$$\log(1 + \bar{\varepsilon}_\ell) = \frac{r_{\ell-1}}{L - \ell + 1}.$$

Since $\varepsilon_c^{(\ell)} \le \bar{\varepsilon}_\ell$,

$$s_\ell \le s_{\ell-1} + \frac{b_1 - s_{\ell-1}}{L - \ell + 1}.$$

Equivalently,

$$b_1 - s_\ell \ge \frac{L - \ell}{L - \ell + 1}(b_1 - s_{\ell-1}).$$

Induction gives $s_L \le b_1$. Proposition 3.1 then implies $\|T_L\|_1 \le \exp(s_L) \le B_1$. □

Certificate failure under a finite cap. Lemma 4.16 is a certificate conditional on every layer reaching its allocated tolerance before the step cap. If some layer fails to reach its assigned tolerance by $t_{\max}$, the algorithm returns a failed certificate rather than a false guarantee. The realized matrix may still be used in training, but the deterministic budget $\|T_L\|_1 \le B_1$ is then not certified by the stopping rule. In that case one should increase $t_{\max}$, reduce the logit dynamic range, or accept a larger budget.

The controller is implementation bookkeeping rather than a separate algorithmic contribution. At layer ℓ, after observing the accumulated log-budget

$$s_{\ell-1} = \sum_{j<\ell} \log(1 + \varepsilon_c^{(j)}),$$

it assigns the next tolerance

$$\bar{\varepsilon}_\ell = \exp\left(\frac{\log B_1 - s_{\ell-1}}{L-\ell+1}\right) - 1. \tag{4.58}$$

The layer then uses the early-stopping rule (4.11). If the cap is reached before $\varepsilon_c^{(\ell)} \le \bar{\varepsilon}_\ell$, the controller reports a failed certificate for that layer; otherwise it updates $s_\ell = s_{\ell-1} + \log(1+\varepsilon_c^{(\ell)})$. Full implementation pseudocode is given in Appendix A.

4.9 Temperature–clipping calibration

The dynamic range is controlled in implementations through logit clipping and temperature. Suppose

$$C_{ij} \in [-R, R], \qquad A_{ij} = \exp(\beta C_{ij}). \tag{4.59}$$

Then

$$\kappa(A) \le \exp(2\beta R), \qquad \frac{\kappa(A)-1}{\kappa(A)+1} \le \tanh(\beta R). \tag{4.60}$$

Therefore the conservative finite-step rule becomes

$$\varepsilon_c^{(2t)} \le \Psi_n\Big(2\beta R \tanh(\beta R)^{2t-1}\Big). \tag{4.61}$$

Remark 4.17 (Temperature–clipping design rule). Fix n, a full-step budget $t_{\max} \ge 1$, and a target column tolerance $\bar{\varepsilon}$. If

$$\Psi_n\Big(2\beta R \tanh(\beta R)^{2t_{\max}-1}\Big) \le \bar{\varepsilon}, \tag{4.62}$$

then every clipped logit matrix $C \in [-R, R]^{n\times n}$ produces, after $t_{\max}$ full column-then-row Sinkhorn steps, a nonnegative mixer satisfying

$$\varepsilon_c^{(2t_{\max})} \le \bar{\varepsilon}.$$

This follows immediately from (4.60), the monotonicity of Ψ_n, and Theorem 4.7. A depth-L backward budget B_1 is certified whenever $\bar{\varepsilon} \le B_1^{1/L} - 1$.

4.10 Certificate-aware training objective

The certificates can also be used as soft training constraints. For a depth-L stack, define

$$\mathcal{C}_{\text{prod}} := \sum_{\ell=1}^{L} \log(1 + \varepsilon_c^{(\ell)}). \tag{4.63}$$

Given a target backward budget B_1, set

$$\mathcal{R}_{\text{budget}} := \left[\mathcal{C}_{\text{prod}} - \log B_1\right]_+^2. \tag{4.64}$$

One may train with

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{cert}}\mathcal{R}_{\text{budget}} + \lambda_\kappa \sum_{\ell=1}^{L} \left[\log \kappa_\ell - \log \kappa_{\max}\right]_+^2. \tag{4.65}$$

The first term penalizes violation of the depth-wise backward budget, while the second prevents the logits from entering a high-dynamic-range regime where many Sinkhorn steps are needed.

If $\mathcal{R}_{\text{budget}} = 0$, then

$$\sum_{\ell=1}^{L} \log(1 + \varepsilon_c^{(\ell)}) \le \log B_1.$$

For entrywise nonnegative mixers, Proposition 3.1 therefore gives

$$\|H^{(L)} \cdots H^{(1)}\|_1 \le B_1. \tag{4.66}$$

Thus the penalty encourages low-budget layers. It does not by itself guarantee stable training; the deterministic guarantee is obtained only from the realized marginal errors and the resulting product certificate.

Calibration summary. See Figure 1 in the Introduction for the mixer-path certificate workflow. Here we provide the formal statement (Theorem 4.15) and a conservative lookup table (Table 1).

Computational note. A full Sinkhorn step (one column and one row normalization) costs $O(n^2)$ time and $O(n^2)$ memory for dense $n \times n$ mixers. The calibration therefore makes the stability–compute trade-off explicit: smaller tolerances or larger dynamic range require larger $t_{\max}$, increasing per-layer cost.

Remark 4.18. The practical motivation for Theorem 4.15 is the Amax behavior reported in (Xie et al., 2025): in an unconstrained baseline, single-layer Amax gains can remain near $O(1)$ while composite backward gains become large at intermediate depth. Our results formalize a deterministic sufficient mechanism: if the implemented finite-step Sinkhorn mixers achieve uniformly small marginal errors, then depth-wise Amax growth is bounded by the corresponding product certificate.

5 Numerical diagnostics for finite-step certificates

The purpose of this section is not to establish a new state-of-the-art architecture or a broad empirical benchmark. Instead, we diagnose the quantities that appear in the deterministic theory: achieved column-marginal error, row defect in implemented arithmetic, the product certificate

$$\prod_{\ell=1}^{L} (1 + \varepsilon_c^{(\ell)}),$$

the exact realized backward mixing gain $\|T_L\|_1$ computed by vector propagation, cap failures in finite-step Sinkhorn, and the gap between worst-case Hilbert bounds and matrix-specific local predictions.

> The experiments test whether the certificate quantities behave as predicted; they are not intended to establish accuracy superiority of mHC or any mHC-style architecture over competing architectures.

Table 1: Conservative full-step budgets $t_{\max}$ sufficient to achieve $\Psi_n\big((\log\kappa)q^{2t_{\max}-1}\big) \le \varepsilon_{\mathrm{tol}}$ (Theorem 4.7).

n	κ	$\varepsilon_{\mathrm{tol}}$	sufficient $t_{\max}$
4	2	5×10^{-3}	3
4	2	1×10^{-3}	4
4	2	5×10^{-4}	4
4	5	5×10^{-3}	8
4	5	1×10^{-3}	10
4	5	5×10^{-4}	11
4	10	5×10^{-3}	16
4	10	1×10^{-3}	20
4	10	5×10^{-4}	21
4	20	5×10^{-3}	32
4	20	1×10^{-3}	40
4	20	5×10^{-4}	43
8	2	5×10^{-3}	3
8	2	1×10^{-3}	4
8	2	5×10^{-4}	4
8	5	5×10^{-3}	8
8	5	1×10^{-3}	10
8	5	5×10^{-4}	11
8	10	5×10^{-3}	16
8	10	1×10^{-3}	20
8	10	5×10^{-4}	22
8	20	5×10^{-3}	32
8	20	1×10^{-3}	40
8	20	5×10^{-4}	44
16	2	5×10^{-3}	3
16	2	1×10^{-3}	4
16	2	5×10^{-4}	4
16	5	5×10^{-3}	8
16	5	1×10^{-3}	10
16	5	5×10^{-4}	11
16	10	5×10^{-3}	16
16	10	1×10^{-3}	20
16	10	5×10^{-4}	22
16	20	5×10^{-3}	33
16	20	1×10^{-3}	41
16	20	5×10^{-4}	44

The current numerical suite uses a CIFAR-10 stress harness and a targeted extra-diagnostic runner. The main run is a real CIFAR-10 training experiment: the softmax and Sinkhorn models are trained on identical augmented minibatches, with paired initialization for all non-mixer parameters and a deterministic per-epoch data order.

Measurement details. The runner records the implemented mixer matrices after the same normalization used in training. Marginal errors are measured as

$$\varepsilon_c^{(\ell)} = \|\mathbf{1}^\top H^{(\ell)} - \mathbf{1}^\top\|_{\infty,\mathrm{vec}}, \qquad \varepsilon_r^{(\ell)} = \|H^{(\ell)}\mathbf{1} - \mathbf{1}\|_{\infty,\mathrm{vec}}.$$

The product certificate is

$$\prod_{\ell=1}^{L}(1+\varepsilon_c^{(\ell)}),$$

and the exact realized backward mixing gain is computed by the vector recursion

$$w_{\ell-1} = H^{(\ell)\top} w_\ell, \qquad \|T_L\|_1 = \|w_0\|_\infty, \qquad w_L = \mathbf{1},$$

which is the diagnostic in Remark 3.2. The same final-epoch weights are also evaluated with fixed $t = t_{\max}$ to separate early-stop behavior from what would be obtained by spending the whole cap.

Stress profile. The main GPU command uses `profile=stress_full`: $L = 128$, $n = 8$ streams, feature dimension $d = 128$, SGD with learning rate 0.02, momentum 0.9, weight decay 5×10^{-4}, Sinkhorn tolerance $\varepsilon_{\mathrm{tol}} = 5 \times 10^{-4}$, and cap $t_{\max} = 80$. The softmax branch uses the same architecture but a row-stochastic softmax mixer. The stress profile deliberately uses larger mixer-logit scale and temperature than the regular sweep so that early stopping, cap failures, and dynamic-range diagnostics are visible.

Local spectral diagnostic. For each Sinkhorn layer, the runner computes the fixed-$t_{\max}$ matrix A^{fix}, uses it as the realized doubly-stochastic approximation P, and reports

$$\rho_* = \sigma_2(P)^2.$$

It also reports the empirical final-step ratio

$$\varepsilon_c^{(2t+2)}/\varepsilon_c^{(2t)}$$

when the denominator is numerically meaningful. This final usable ratio is a diagnostic, but near the early-stopping tolerance or numerical floor it need not estimate the asymptotic local rate. A direct comparison with the local spectral predictor in Theorem 4.10 would require recording the pre-tolerance decay curve.

Generated tables. The GPU run writes final-epoch diagnostics for dynamic range, early stopping, exact gain, the product certificate, ρ_*, final usable ratio, and cap failure. In the repository package, the small `results-cpu-smoke` run is only an execution sanity check and should not be used as a paper result. The reported tables were generated from `results-stressed-cap80/logs/kappa_tstop_table.tex` using the reproducibility command below.

Model	med. κ_{obs}	std	med. $\kappa_{\mathrm{obs}}^{\mathrm{fix}}$	std
softmax	8556629.04	840682.39	—	—
sinkhorn (early)	14489.91	1315.73	14487.00	1324.17

Table 2: Final-epoch dynamic-range diagnostics (profile: `stress_full`). A^{fix} denotes the oracle evaluation with fixed $t = t_{\max}$ using the same trained weights. For the row-softmax baseline, κ_{obs} is a native pre-softmax dynamic-range diagnostic, not the clipped positive-matrix dynamic range used in the Sinkhorn certificate.

Model	med. t_{stop}	hit	$\|c(A) - \mathbf{1}\|_\infty$	$\|c(A^{\mathrm{fix}}) - \mathbf{1}\|_\infty$
sinkhorn (early)	14.00	1.00	0.000499±1e-06	0.000187±0.0001

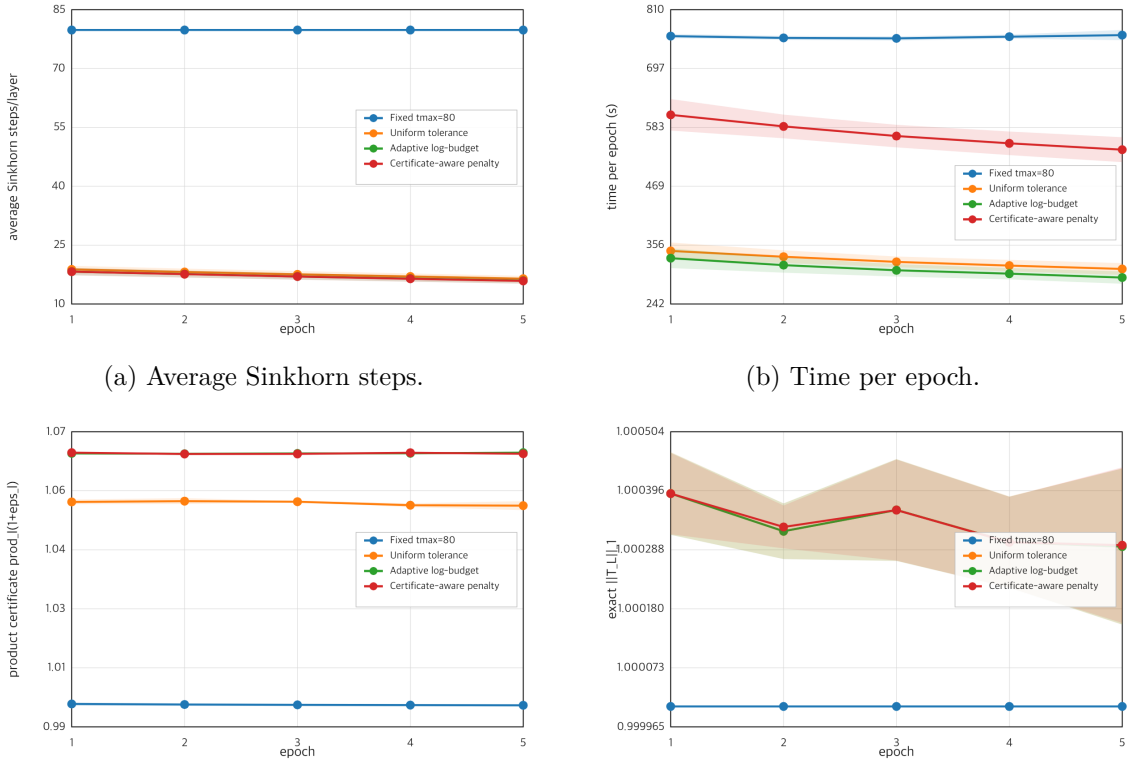
Table 3: Final-epoch finite-step Sinkhorn controller diagnostics. The early-stop tolerance is $\varepsilon = 0.0005$ and the cap is $t_{\max} = 80$. The fixed column defect is measured after spending the full cap using the same trained weights.

Focused controller ablation. To test the controller in Lemma 4.16 and the soft penalty (4.65), we reran the same CIFAR-10 stress setting with only the Sinkhorn stopping/control policy changed. Table 5 compares fixed cap, uniform early stopping, adaptive log-budget stopping, and an adaptive policy trained with the certificate-aware penalty. This is a targeted diagnostic ablation, not a large-scale benchmark.

The fixed-cap policy is much more expensive and produces a certificate far tighter than the prescribed budget. Uniform tolerance already recovers essentially the same accuracy and exact realized backward gain using about one fifth of the Sinkhorn iterations. The adaptive log-budget policy spends the intended product budget almost exactly and gives an additional compute reduction relative to uniform tolerance: about 3.1% fewer Sinkhorn iterations and about 4.8% lower epoch time in this run. The certificate-aware penalty row preserves the same accuracy, exact realized gain, cap-failure rate, and ρ_*, but is slower in this implementation because it computes extra differentiable certificate terms during training. Thus the penalty row is evidence of feasibility, while the main compute-saving mechanism is the adaptive stopping controller. Figure 2 shows the corresponding three-seed trajectories.

Temperature–clipping grid. Table 6 and Figure 3 test the design inequality (4.61) on random clipped logit matrices with $n = 8$. The measured column errors are always below the finite-precision certificate when the certificate is above the numerical floor. The grid also shows why the temperature–clipping rule should be read as a conservative design certificate rather than a sharp predictor: for large βR, the worst-case certificate is nearly vacuous even though the measured errors decrease rapidly with $t_{\max}$.

Local-predictor curve. For trained adaptive Sinkhorn matrices, the extra runner records the full curve $\varepsilon_c^{(2t)}$ for $t = 1, \dots, 80$ and compares it with the κ-only bound, the matrix-adaptive Hilbert bound, and the first- and second-order predictors from Subsection 4.5. Across the six recorded layer/seed curves, the median absolute $\log_{10}$ error is 0.26 for the first-order predictor and 0.19 for the second-order predictor, while the two global Hilbert



(a) Average Sinkhorn steps.

(b) Time per epoch.

(c) Product certificate.

(d) Exact propagated gain.

Figure 2: Focused controller-ablation trajectories over five epochs. Curves and standard-deviation bands are computed only over seeds $0, 1, 2$.

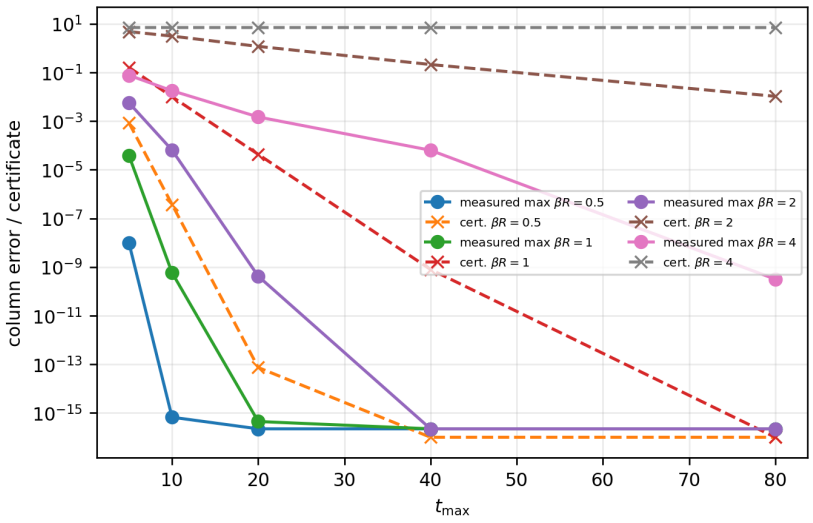


Figure 3: Temperature–clipping diagnostic for the conservative design rule (4.61). Solid curves are measured maximum column errors over random clipped logits; dashed curves are the corresponding worst-case certificates.

Model	cert.	exact $\|T_L\|_1$	exact/cert.
softmax	3.61e+47±5e+47	5.72±2	—
sinkhorn (early)	1.05±0.0009	1±8e-05	0.951

Table 4: Final-epoch stack certificate diagnostics. The exact backward gain is computed by vector propagation, while the certificate is $\prod_\ell(1+\varepsilon_c^{(\ell)})$. For Sinkhorn, the local diagnostic is $\rho_* = 0.631 \pm 0.01$, the final usable ratio is 1±0, and the cap-failure rate is 0. The final usable ratio is a finite-run diagnostic, not an asymptotic local-rate estimate near tolerance or the numerical floor.

Policy	Acc.	Time/epoch	Avg. steps	Product cert.	Exact $\|T_L\|_1$	Cap fail.	ρ_*
Fixed $t_{\max} = 80$	0.5835 ± 0.0373	758.0 ± 6.1s	80.00 ± 0.00	1.00038 ± 0.00032	1.000002 ± 0.00000007	$0.00 \pm 0.00\%$	0.622 ± 0.013
Uniform tolerance	0.5835 ± 0.0369	325.4 ± 17.6s	16.36 ± 0.71	1.05213 ± 0.00117	1.000294 ± 0.00014273	$0.26 \pm 0.37\%$	0.622 ± 0.013
Adaptive log-budget	0.5809 ± 0.0342	309.6 ± 19.1s	15.86 ± 0.73	1.06587 ± 0.00014	1.000294 ± 0.00014248	$0.26 \pm 0.37\%$	0.622 ± 0.013
Certificate-aware penalty	0.5842 ± 0.0380	570.2 ± 34.2s	15.80 ± 0.72	1.06558 ± 0.00002	1.000297 ± 0.00014210	$0.26 \pm 0.37\%$	0.621 ± 0.013

Table 5: Focused controller ablation under the CIFAR-10 stress setting. The adaptive and certificate-aware policies use the shared budget $B_1 = (1 + 5 \cdot 10^{-4})^{128} = 1.066075$. Statistics are computed over seeds $0, 1, 2$.

certificates are more than five orders of magnitude conservative at the median. Figure 4 shows a representative trained layer.

Observed dynamic range. The final available median observed dynamic range across the Sinkhorn policies in the focused run is $\kappa_{\text{obs}} \approx 1.27 \times 10^4$. This does not contradict the small-κ lookup table in Appendix D: the table is an illustrative conservative design example, whereas κ_{obs} is a realized diagnostic of trained clipped logits. Large observed dynamic range mainly explains why the κ-only Hilbert bound is conservative, and why the numerical section reports measured marginal errors, product certificates, exact propagated gains, and local predictors next to the design rule.

βR	$t_{\max}$	max measured ε_c	certificate	max/cert.
0.5	5	$9.79e-09$	$8.41e-04$	$1.16e-05$
0.5	10	$6.66e-16$	$3.74e-07$	$1.78e-09$
0.5	20	$2.22e-16$	$7.36e-14$	0.00302
0.5	40	$2.22e-16$	$< 10^{-15}$	—
0.5	80	$2.22e-16$	$< 10^{-15}$	—
1.0	5	$3.81e-05$	0.161	$2.37e-04$
1.0	10	$5.82e-10$	0.00995	$5.85e-08$
1.0	20	$4.44e-16$	$4.27e-05$	$1.04e-11$
1.0	40	$2.22e-16$	$7.93e-10$	$2.80e-07$
1.0	80	$2.22e-16$	$< 10^{-15}$	—
2.0	5	0.00563	4.74	0.00119
2.0	10	$6.38e-05$	3.1	$2.06e-05$
2.0	20	$4.18e-10$	1.17	$3.57e-10$
2.0	40	$2.22e-16$	0.21	$1.06e-15$
2.0	80	$2.22e-16$	0.0104	$2.14e-14$
4.0	5	0.0763	6.98	0.0109
4.0	10	0.0176	6.98	0.00252
4.0	20	0.00147	6.98	$2.11e-04$
4.0	40	$6.28e-05$	6.97	$9.01e-06$
4.0	80	$3.16e-10$	6.96	$4.54e-11$

Table 6: Temperature–clipping grid for the design rule $\Psi_n(2\beta R \tanh(\beta R)^{2t_{\max}-1})$. Measured errors are maxima over 64 random clipped logit matrices with $n = 8$. Entries shown as $< 10^{-15}$ are below the numerical floor of this double-precision diagnostic.

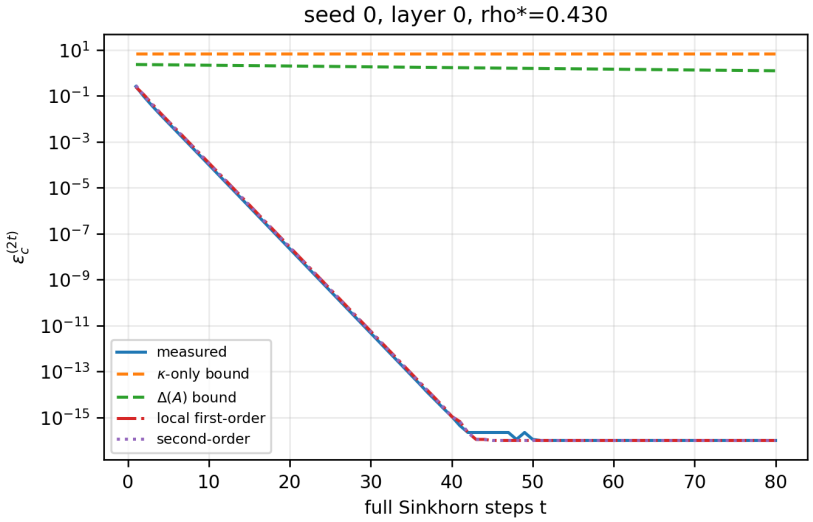
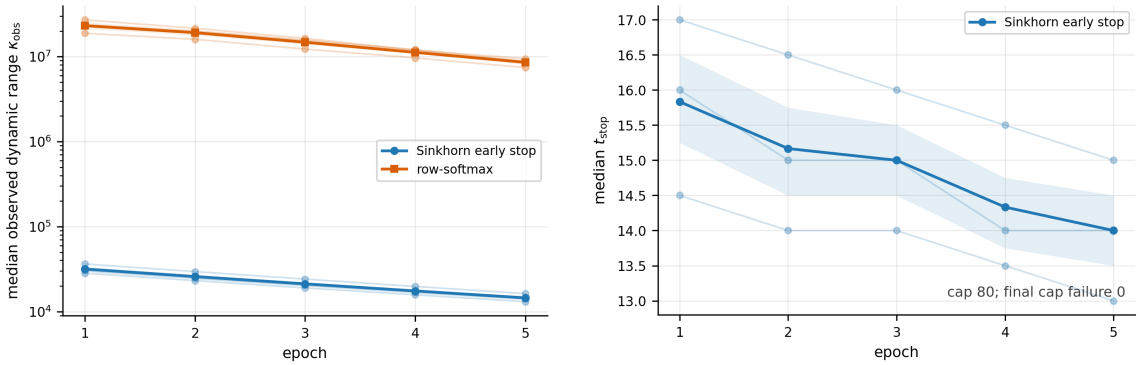


Figure 4: Representative local-predictor comparison on a trained adaptive Sinkhorn layer. The local first- and second-order predictors track the measured Sinkhorn decay closely, while the safe Hilbert certificates remain conservative.

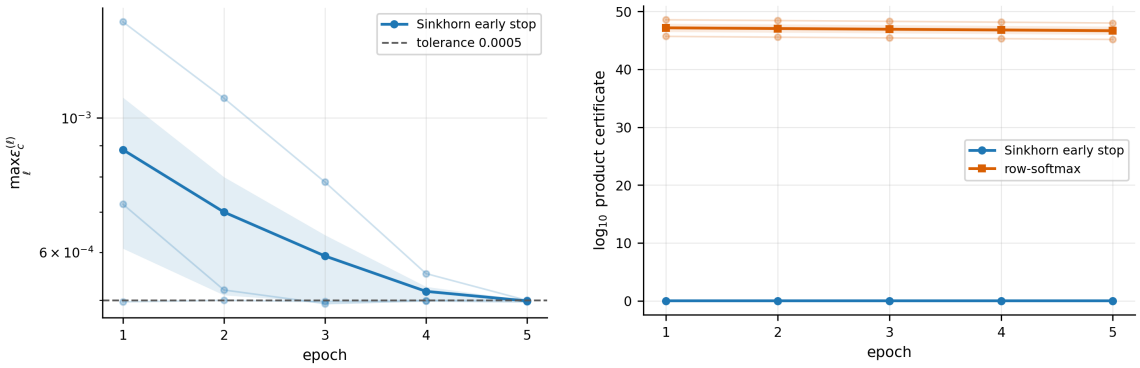
Generated figures. The real training log also supports the paper figures below. The plotted quantities are epoch summaries across the GPU seeds rather than synthetic or random-matrix diagnostics; the script `make_paper_plots.py` regenerates them directly from `epoch_logs.csv`. Auxiliary plots for test accuracy and the empirical final-step ratio are written as checks, but the paper uses only the controller diagnostics shown here.



(a) Observed dynamic range, log scale.

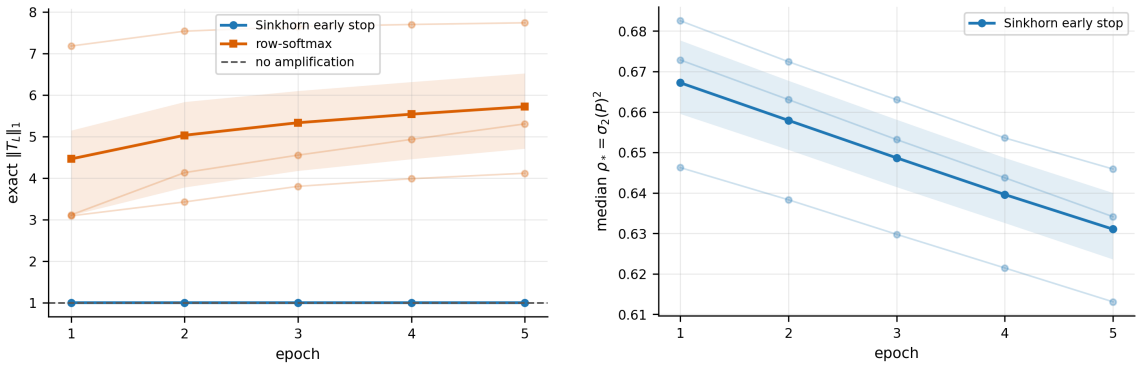
(b) Early-stop step counts.

Figure 5: CIFAR-10 stress diagnostics for the finite-step Sinkhorn controller.



(a) Maximum column-marginal error across layers. (b) Log-scale product certificate $\prod_\ell (1 + \varepsilon_c^{(\ell)})$.

Figure 6: Measured marginal error and the corresponding product certificate.



(a) Exact backward gain by vector propagation. (b) Median local spectral rate ρ_*.

Figure 7: Exact realized gain and the local spectral predictor recorded during training.

Scope of the reported experiments. The current code validates the fixed finite-step and early-stop controller diagnostics and reports targeted diagnostics for adaptive global-budget stopping, the certificate-aware penalty, the temperature–clipping rule, and the local spectral predictors. It does not report a broad dataset or architecture benchmark; those larger comparisons are left out rather than reported as unsupported numerical values.

6 Discussion and practical guidance

6.1 Practical guidance

We summarize practical guidance for designing finite-step Sinkhorn mixers when the objective is to control Amax amplification across depth:

- **Nonnegativity is required for worst-case Amax certificates.** The signed counterexample from Section 3.1 shows that marginal constraints alone cannot prevent arbitrarily large $\|\cdot\|_\infty$, $\|\cdot\|_1$, or $\|\cdot\|_2$ amplification when signs are allowed.
- **Depth-L budgets translate into per-layer tolerances.** For small errors, a composite budget $\|T_L\|_1 \le B_1$ corresponds to $\sum_{\ell=1}^{L} \varepsilon_c^{(\ell)} \lesssim \log B_1$, and a simple uniform choice is $\varepsilon_c \approx \log(B_1)/L$ (see Remark 3.5).
- **Given a target tolerance, $(\kappa, t_{\max})$ can be selected via the finite-step rule.** Given a target ε_c, the corrected finite-step rule (Theorem 4.7) provides a sufficient $t_{\max}$ through $\Psi_n\big((\log\kappa)q^{2t_{\max}-1}\big) \le \varepsilon_c$. Smaller κ improves contraction (fewer steps) but restricts routing selectivity (see Remark 4.14).
- **For prediction, use the local spectral rate.** The κ-only bound is a worst-case certificate. When a numerical prediction of the marginal-error curve is needed, compute the Sinkhorn limit P and use the local rate $\sigma_2(P)^2$ and predictor (4.49).
- **For training, use a global-budget controller rather than a fixed tolerance whenever possible.** The adaptive rule in Lemma 4.16 spends the remaining log-budget across the remaining layers, and the exact vector propagation certificate in Remark 3.2 measures the realized gain.
- **For implementation, calibrate temperature and clipping jointly.** Remark 4.17 converts $(\beta, R, t_{\max})$ directly into a conservative tolerance. Large βR increases selectivity but slows the worst-case Sinkhorn certificate.

Remark 6.1 (Transfer beyond mHC: attention and message passing). The results in Sections 3–4 apply verbatim to any architecture whose depth-wise propagation includes a product of *entrywise nonnegative* approximately bistochastic mixing matrices. This includes, for example, (i) Sinkhorn-normalized attention maps (e.g., doubly-stochastic attention) when the normalization is performed on the relevant (unmasked) support, and (ii) reweighted graph message-passing operators that are constrained to be approximately doubly stochastic. These guarantees assume strict positivity on the support being normalized; masked attention or sparse normalization can violate the standing assumptions of Section 4 and requires separate treatment (e.g., restricting to the unmasked block).

In these settings, Proposition 3.1 certifies worst-case amplification of the *mixing sub-operator*. Full-block stability still requires coupling this certificate with bounds on the remaining components (e.g., projections/MLPs/normalization), as recorded in Appendix E.

6.2 Limitations

Our results provide *sufficient* stability certificates tailored to the Amax diagnostic and the finite-step Sinkhorn mechanism studied here. The analysis is intentionally modular, and several aspects are outside the present scope.

Worst-case nature and conservatism. Our deterministic certificates are worst-case and can be conservative. The product bound in Proposition 3.1 may be loose when mixers have additional structure, although the realized backward gain can be computed exactly by Remark 3.2. Similarly, the Hilbert-metric constants in Theorem 4.7 are safe global certificates. The local envelope in Theorem 4.11 is much sharper for matrices whose logarithmic Sinkhorn orbit lies near the limit, but it is still local. The first- and second-order predictors explain typical numerical behavior; they are not uniform worst-case guarantees.

Finite-step Sinkhorn constants. The step-count rule uses Hilbert-metric contraction bounds. The matrix-adaptive diameter $\Delta(A)$ and initialization δ_0 already improve the κ-only estimate, but they remain global certificates. The derivative envelope based on ρ_R and λ_R is sharper and non-asymptotic on a verified local ball. The first-order rate $\sigma_2(P)^2$ gives the late-stage slope, while the second-order expansion improves the first few finite steps.

Nonlinear blocks and full-network training dynamics. The influence of the nonlinear block F enters only through Jacobian norm bounds such as Theorem E.1 and Corollary E.2. We do not model how training shapes $\widetilde{H_{\mathrm{res}}}^{(\ell)}$ (and hence κ or marginal errors), nor do we analyze how the certificates interact with optimization, data distribution, or generalization.

Architectural and implementation choices. Our statements track the streams-as-rows convention and a specific normalization order for finite-step Sinkhorn. Other conventions (e.g. streams-as-columns) or implementation details (e.g. damping, mixed precision, sparse mixers) require straightforward but explicit translation, and can affect which marginal error is nontrivial at the output.

Connecting worst-case certificates to learned behavior would require sharper finite-step constants or additional structure beyond marginal errors.

6.3 Future work

Three directions would improve practicality: (i) turning the local derivative envelope into cheap certified code for each batch/layer, including reliable upper bounds for ρ_R and λ_R; (ii) expanding the controller and certificate-penalty ablations across datasets and architectures, including more efficient implementations of the differentiable penalty; and (iii) deriving tighter full-network bounds that couple mixer certificates to the Jacobians of the nonlinear blocks, rather than using a uniform L_F surrogate.

7 Conclusion

We developed deterministic finite-step certificates for nonnegative Sinkhorn-normalized stream mixers. The central point is that, for nonnegative matrices, marginal errors are not merely normalization diagnostics: they are induced-norm certificates. In the `column-then-row` Sinkhorn order, the output row sums are exact in ideal arithmetic and the remaining output column defect gives a backward Amax budget for the depth-wise mixer product. Hilbert projective contraction turns logit dynamic range and finite Sinkhorn step count into a conservative design-time tolerance, while matrix-specific local analysis explains why realized convergence is often much faster than the worst-case bound.

The resulting theory supports adaptive early stopping, budget tracking, and exact realized gain diagnostics for the mixing pathway. The focused ablations show that, at the same depth-wise product budget, adaptive log-budget stopping matches the uniform tolerance rule in accuracy and exact backward gain while using modestly fewer Sinkhorn steps. The certificate-aware penalty preserves the same stability diagnostics but is slower in the current implementation, so it should be read as a feasibility/control diagnostic rather than as a speedup. Extending these mixer-path certificates to full network-level guarantees requires additional control of the nonlinear residual branches, and extending the finite-step theory to masked or sparse supports remains an important direction for future work.

Reproducibility Statement

We provide scripts to regenerate all figures and tables, with fixed random seeds and commands referenced by figure/table number. Because numerical behavior depends on implementation details, we report the Sinkhorn dtype, the positivity floor $\varepsilon_{\text{floor}}$, clipping/temperature settings, and the early-stopping tolerance; these quantities directly affect measured marginal errors and the resulting certificates.

Code: The code, raw CSV/JSON logs, regeneration scripts, fixed seeds, and a CPU smoke test are available at

https://github.com/MaKosmakov/finite-step-sinkhorn-certificates

Exact regeneration commands.

```
# CPU smoke test on real CIFAR-10 data; validates code paths only.
MPLCONFIGDIR=/tmp/mplconfig python train_cifar10_mhc.py \
  --profile baseline --epochs 1 --seeds 0 \
  --num_workers 0 --batch_size 64 \
  --n_streams 4 --d 32 --depth 4 \
  --sinkhorn_t_max 8 --sinkhorn_eps_tol 5e-4 \
  --max_train_batches 1 --max_eval_batches 1 \
  --data_root results-stressed/data \
  --out_dir results-cpu-smoke

# Full CIFAR-10 stress run used for paper diagnostics.
python train_cifar10_mhc.py \
  --profile stress_full --epochs 5 --seeds 0 1 2 \
  --batch_size 256 --num_workers 4 \
```

```
  --data_root results-stressed/data \
  --out_dir results-stressed-cap80 \
  --amp

# Extra diagnostics: temperature-clipping grid and local-predictor curves.
python paper_extra_experiments_kaggle.py

# Focused controller ablation used for Table 5 and Figure 2.
python paper_extra_experiments_3seeds_5epochs_kaggle.py
python paper-extra-experiments/regenerate_focused_ablation_3seed.py \
  paper-extra-experiments-3seeds-5epochs/logs/focused_ablation_epoch_logs.csv \
  paper-extra-experiments
```

The repository command manifest gives the full, paper-indexed list of commands and environment details.

A Implementation pseudocode for certificate bookkeeping

The following listings are not separate algorithmic contributions. They specify the bookkeeping used to evaluate the finite-step certificates in implementation.

```python
def sinkhorn_col_row(M, t_max, eps_tol, floor=1e-12):
    # M: positive matrix (e.g., exp(beta * logits))
    X = np.maximum(M.copy(), floor)
    tiny = np.finfo(X.dtype).tiny

    for t in range(1, t_max + 1):
        # Full step: column then row normalization
        X /= np.maximum(X.sum(axis=0, keepdims=True), tiny)  # columns
        X /= np.maximum(X.sum(axis=1, keepdims=True), tiny)  # rows

        # column-marginal error at full-step output S^{(2t)}
        eps_c = np.max(np.abs(X.sum(axis=0) - 1.0))
        if eps_c <= eps_tol:
            return X, t  # t is the number of full steps used

    return X, t_max
```

Listing 1: Early stopping for `column-then-row` Sinkhorn to a column-marginal tolerance.

```python
def sinkhorn_global_budget(logits_list, B1, t_cap, eps_floor=1e-12):
    """Return mixers and a realized product certificate."""
    L = len(logits_list)
    b = math.log(B1)
    s = 0.0
    H_list, eps_list, step_list = [], [], []
    budget_failed = False

    for ell, C in enumerate(logits_list):
        remaining_layers = L - ell
        eps_target = math.exp((b - s) / remaining_layers) - 1.0

        A = torch.exp(C).clamp_min(eps_floor)
```

```
        H, eps_c, steps = sinkhorn_until_column_tol(
            A, eps_target=eps_target, t_cap=t_cap, eps_floor=eps_floor
        )

        H_list.append(H)
        eps_list.append(eps_c)
        step_list.append(steps)

        if float(eps_c) > eps_target:
            budget_failed = True
        s += math.log1p(float(eps_c))

    product_certificate = math.exp(s)
    return H_list, eps_list, step_list, product_certificate, budget_failed
```

Listing 2: Adaptive global-budget finite-step Sinkhorn early stopping.

B Auxiliary proof details

Proof of Lemma 4.4. Let $a := \max_i u_i$, $b := \min_i u_i$, and $r := a/b = e^{\delta} \geq 1$. The mean constraint gives $b \leq 1 \leq a$. For the upper deviation, fixing the ratio $a/b = r$ and the mean, the largest possible value of a occurs when one entry is a and the remaining $n-1$ entries are b. Hence

$$a \leq \frac{nr}{r+n-1}, \qquad a - 1 \leq \frac{(n-1)(r-1)}{r+n-1}.$$

For the lower deviation, the smallest possible value of b occurs when one entry is b and the remaining $n-1$ entries are a. Thus

$$b \geq \frac{n}{1+(n-1)r}, \qquad 1 - b \leq \frac{(n-1)(r-1)}{1+(n-1)r}.$$

Since $1 + (n-1)r \geq r + n - 1$ for $n \geq 2$ and $r \geq 1$, the lower-tail bound is no larger than the upper-tail bound. Therefore

$$\|u - \mathbf{1}\|_{\infty,\mathrm{vec}} \leq \frac{(n-1)(r-1)}{r+n-1} = \Psi_n(\delta).$$

Solving $\Psi_n(\delta) \leq \varepsilon$ for $r = e^{\delta}$ gives (4.15).

Proof of Lemma 4.5. Since $z = \Pi \log u$ and $n^{-1}\sum_i u_i = 1$, there is a unique scalar normalization such that

$$u_i = \frac{ne^{z_i}}{\sum_{k=1}^{n} e^{z_k}}.$$

We first maximize a single coordinate. Under the constraints $\sum_i z_i = 0$ and $\|z\|_2 \leq \eta$, the largest possible value of one coordinate is

$$a = \sqrt{\frac{n-1}{n}}\,\eta,$$

attained when the remaining coordinates all equal $-a/(n-1)$. Indeed, if $z_i = a$ is fixed, the sum of the other coordinates is $-a$; Jensen's inequality gives

$$\sum_{k\neq i} e^{z_k} \geq (n-1)e^{-a/(n-1)},$$

with equality exactly in the equal-tail configuration. Therefore

$$u_i \leq \frac{ne^a}{e^a + (n-1)e^{-a/(n-1)}}.$$

This gives the upper-tail deviation $\Gamma_n(\eta)$.

It remains to check that the lower-tail deviation is not larger. Put

$$r := \exp\left(\frac{n}{n-1}a\right) \geq 1.$$

For the upper-tail extremizer, the maximal coordinate is

$$U = \frac{nr}{r+n-1}.$$

For the lower-tail extremizer, where one coordinate is $-a$ and the other $n-1$ coordinates are $a/(n-1)$, the minimal coordinate is

$$L = \frac{n}{1+(n-1)r}.$$

The upper and lower deviations are $U-1$ and $1-L$. A direct calculation gives

$$(U-1) - (1-L) = U + L - 2 = \frac{(n-1)(n-2)(r-1)^2}{(r+n-1)(1+(n-1)r)} \geq 0. \tag{B.1}$$

Thus the one-spike upper-tail value controls the full ℓ_∞ deviation. Expanding (4.17) at $\eta = 0$ gives (4.18).

C Second-order local predictor formulas

The second-order diagnostic predictor keeps the quadratic terms in the logarithmic Sinkhorn map. With $K = PP^\top$, define

$$Q_F(x) := -P((P^\top x)^{\odot 2}) + \frac{1}{2}K(x^{\odot 2}) + \frac{1}{2}(Kx)^{\odot 2}, \tag{C.1}$$

$$Q_L(x) := \frac{1}{2}P^\top(x^{\odot 2}) - \frac{1}{2}(P^\top x)^{\odot 2}. \tag{C.2}$$

Then

$$\mathcal{F}(x) = Kx + \Pi Q_F(x) + O(\|x\|^3), \qquad \mathcal{L}(x) = \Pi P^\top x + \Pi Q_L(x) + O(\|x\|^3). \tag{C.3}$$

Set

$$x_t^{[1]} := K^t x_0, \qquad x_t^{[2]} := K^t x_0 + \sum_{s=0}^{t-1} K^{t-1-s} \Pi Q_F(K^s x_0). \tag{C.4}$$

The second-order centered log-column predictor is

$$\ell_t^{[2]} := \Pi\Big[P^\top x_t^{[2]} + Q_L(x_t^{[1]}) - P^\top x_{t-1}^{[2]} - Q_L(x_{t-1}^{[1]})\Big]. \tag{C.5}$$

Finally define

$$\widehat{v}_t^{[2]} := \frac{n \exp(\ell_t^{[2]})}{\sum_{j=1}^n \exp((\ell_t^{[2]})_j)}, \qquad \widehat{\varepsilon}_{c,2}^{(2t)} := \|\widehat{v}_t^{[2]} - \mathbf{1}\|_{\infty,\mathrm{vec}}. \tag{C.6}$$

For each fixed t,

$$z_t = \ell_t^{[2]} + O_t(\|x_0\|^3). \tag{C.7}$$

Thus (C.6) is the natural finite-step prediction to compare with early numerical points, while (4.49) gives the asymptotic slope.

D One explicit sufficient instantiation of the finite-step rule

This appendix provides an explicit instantiation of the corrected step-count rule in Theorem 4.7.

Assume the positive pre-scaling matrix $S^{(0)}$ has dynamic range

$$\kappa := \frac{\max_{i,j} S_{ij}^{(0)}}{\min_{i,j} S_{ij}^{(0)}}. \tag{D.1}$$

By Birkhoff–Bushell contraction theory for positive maps in the Hilbert projective metric Birkhoff (1957); Bushell (1973), a conservative contraction bound is

$$q = \frac{\kappa - 1}{\kappa + 1}. \tag{D.2}$$

The corrected output column-marginal estimate in the `column-then-row` order is

$$\varepsilon_c^{(2t)} \le \Psi_n\Big((\log \kappa) q^{2t-1}\Big), \qquad t \ge 1. \tag{D.3}$$

Thus it suffices to choose $t_{\max}$ so that

$$t_{\max} \ge \frac{1}{2}\left[1 + \frac{\log\big((\log \kappa)/\Psi_n^{-1}(\bar{\varepsilon})\big)}{\log(1/q)}\right], \tag{D.4}$$

rounded up.

quantity	value
n	8
κ	10
$q = (\kappa - 1)/(\kappa + 1)$	0.818182
$\log \kappa$	2.30259
$\Psi_n^{-1}(10^{-3})$	0.00114237
sufficient $t_{\max}$ from (D.4)	20
$\Psi_n\big((\log \kappa) q^{2t_{\max}-1}\big)$ at $t_{\max} = 20$	8.045×10^{-4}

Table 7: Numerical evaluation of the corrected explicit design-law bound (Theorem 4.7) for $(n, \kappa, \bar{\varepsilon}) = (8, 10, 10^{-3})$.

Numerical examples for $n = 4$. For target marginal error $\bar{\varepsilon} = 10^{-3}$ and $n = 4$, the table below reports $q = (\kappa - 1)/(\kappa + 1)$, $\log \kappa$, and the sufficient $t_{\max}$ obtained from Equation (D.4) with Ψ_n^{-1} from Equation (4.15).

κ	$q = (\kappa - 1)/(\kappa + 1)$	$\log \kappa$	sufficient $t_{\max}$
2	0.333	0.693	4
5	0.667	1.609	10
10	0.818	2.303	20

An explicit evaluation of the corrected design-law bound for a representative $(n, \kappa, \bar{\varepsilon})$ triple is shown in Table 7.

E Full-layer Jacobian Amax bounds

The mixer-product certificates in Section 3 isolate the contribution of stream mixing. This section provides induced-norm bounds for the *full* layer Jacobian of (2.1) in the streams-as-rows convention, under a consistent vectorization.

Let $X \in \mathbb{R}^{n \times C}$ and define the row-major vectorization

$$\mathrm{vec}_r(X) := \mathrm{vec}(X^\top) \in \mathbb{R}^{nC}, \tag{E.1}$$

where vec is the standard column-stacking operator. With this convention one has, whenever products are well-defined,

$$\mathrm{vec}_r(AXB) = (A \otimes B^\top)\, \mathrm{vec}_r(X), \tag{E.2}$$

and in particular

$$\mathrm{vec}_r(HX) = (H \otimes I_C)\, \mathrm{vec}_r(X). \tag{E.3}$$

Layer Jacobian and induced-norm bounds Fix a layer ℓ. Let $H_{\mathrm{res}}^{(\ell)} \in \mathbb{R}^{n \times n}$ be the residual mixer and $H_{\mathrm{pre}}^{(\ell)}, H_{\mathrm{post}}^{(\ell)} \in \mathbb{R}^{1 \times n}$ the pre/post aggregation row vectors. Let $F^{(\ell)} : \mathbb{R}^C \to \mathbb{R}^C$ denote the feature map applied to the aggregated feature vector $u := H_{\mathrm{pre}}^{(\ell)} X \in \mathbb{R}^C$, and let $J_F^{(\ell)} \in \mathbb{R}^{C \times C}$ be the Jacobian of $F^{(\ell)}$ with respect to u.

The layer update is

$$X \mapsto H_{\text{res}}^{(\ell)} X + H_{\text{post}}^{(\ell)\top} F^{(\ell)}(H_{\text{pre}}^{(\ell)} X). \tag{E.4}$$

Applying Equation (E.2) and the chain rule yields the Jacobian (with respect to $\text{vec}_r(X)$):

$$J^{(\ell)} = H_{\text{res}}^{(\ell)} \otimes I_C + \left(H_{\text{post}}^{(\ell)\top} \otimes I_C\right) J_F^{(\ell)} \left(H_{\text{pre}}^{(\ell)} \otimes I_C\right). \tag{E.5}$$

Theorem E.1 (Layer Jacobian induced-norm bounds)**.** *With the notation above,*

$$\left\|J^{(\ell)}\right\|_\infty \le \left\|H_{\text{res}}^{(\ell)}\right\|_\infty + \left\|H_{\text{post}}^{(\ell)}\right\|_{\infty,\text{vec}} \left\|J_F^{(\ell)}\right\|_\infty \left\|H_{\text{pre}}^{(\ell)}\right\|_{1,\text{vec}}. \tag{E.6}$$

Similarly,

$$\left\|J^{(\ell)}\right\|_1 \le \left\|H_{\text{res}}^{(\ell)}\right\|_1 + \left\|H_{\text{post}}^{(\ell)}\right\|_{1,\text{vec}} \left\|J_F^{(\ell)}\right\|_1 \left\|H_{\text{pre}}^{(\ell)}\right\|_{\infty,\text{vec}}. \tag{E.7}$$

Proof. Using Equation (E.5) and submultiplicativity of induced norms,

$$\left\|J^{(\ell)}\right\|_\infty \le \left\|H_{\text{res}}^{(\ell)} \otimes I_C\right\|_\infty + \left\|H_{\text{post}}^{(\ell)\top} \otimes I_C\right\|_\infty \left\|J_F^{(\ell)}\right\|_\infty \left\|H_{\text{pre}}^{(\ell)} \otimes I_C\right\|_\infty. \tag{E.8}$$

Since $\|A \otimes I_C\|_\infty = \|A\|_\infty$ for any A, this becomes

$$\left\|J^{(\ell)}\right\|_\infty \le \left\|H_{\text{res}}^{(\ell)}\right\|_\infty + \left\|H_{\text{post}}^{(\ell)\top}\right\|_\infty \left\|J_F^{(\ell)}\right\|_\infty \left\|H_{\text{pre}}^{(\ell)}\right\|_\infty. \tag{E.9}$$

Finally, because $H_{\text{post}}^{(\ell)\top}$ is a column vector, $\|H_{\text{post}}^{(\ell)\top}\|_\infty = \|H_{\text{post}}^{(\ell)}\|_{\infty,\text{vec}}$, and because $H_{\text{pre}}^{(\ell)}$ is a row vector, $\|H_{\text{pre}}^{(\ell)}\|_\infty = \|H_{\text{pre}}^{(\ell)}\|_{1,\text{vec}}$, yielding Equation (E.6). The $\|\cdot\|_1$ bound Equation (E.7) follows analogously. □

Corollary E.2 (End-to-end Jacobian certification)**.** *Assume each residual mixer is entrywise nonnegative and $(\varepsilon_r^{(\ell)}, \varepsilon_c^{(\ell)})$-doubly stochastic so that $\left\|H_{\text{res}}^{(\ell)}\right\|_\infty \le 1 + \varepsilon_r^{(\ell)}$. If, in addition, there exist constants $b_{\text{pre}}, b_{\text{post}}, L_F \ge 0$ such that for all layers ℓ,*

$$\left\|H_{\text{pre}}^{(\ell)}\right\|_{1,\text{vec}} \le b_{\text{pre}}, \qquad \left\|H_{\text{post}}^{(\ell)}\right\|_{\infty,\text{vec}} \le b_{\text{post}}, \qquad \left\|J_F^{(\ell)}\right\|_\infty \le L_F, \tag{E.10}$$

then

$$\left\|J^{(\ell)}\right\|_\infty \le (1 + \varepsilon_r^{(\ell)}) + b_{\text{post}}\, L_F\, b_{\text{pre}}. \tag{E.11}$$

Thus

$$\left\|J^{(L)} \cdots J^{(1)}\right\|_\infty \le \prod_{\ell=1}^{L} \left((1 + \varepsilon_r^{(\ell)}) + b_{\text{post}}\, L_F\, b_{\text{pre}}\right). \tag{E.12}$$

Remark E.3 (On bounding L_F). The constant L_F depends on the block F. For pointwise MLP blocks, one may bound L_F by products of induced operator norms of the affine maps times activation Lipschitz constants (in $\|\cdot\|_\infty$). For attention-style blocks, L_F depends additionally on projection norms and the (data-dependent) attention map; in such settings, we treat L_F as an architecture- and data-dependent constant and report explicitly how the certificate scales with it.

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